

# Treatment Assignment Design Evaluation for Spatial CRTs

Simulation Study with Heterogeneous Incidence — Results Summary

Andrew Walther

2026-09-24

## Contents

|          |                                                |           |
|----------|------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                            | <b>2</b>  |
| <b>2</b> | <b>Mathematical Framework</b>                  | <b>2</b>  |
| 2.1      | Study Region and Spatial Structure . . . . .   | 2         |
| 2.2      | Outcome Data Generating Process . . . . .      | 2         |
| 2.3      | Incidence Generation . . . . .                 | 3         |
| 2.4      | Treatment Assignment Designs . . . . .         | 3         |
| 2.5      | Estimation Methods . . . . .                   | 4         |
| <b>3</b> | <b>Simulation Algorithm</b>                    | <b>4</b>  |
| 3.1      | Parameter Grid . . . . .                       | 4         |
| 3.2      | Resampling Structure . . . . .                 | 5         |
| 3.3      | Evaluation Metrics . . . . .                   | 5         |
| 3.4      | Per-Draw Reproducibility . . . . .             | 5         |
| <b>4</b> | <b>Results: Difference in Means (DIM)</b>      | <b>5</b>  |
| 4.1      | Overall Summary (All DIM Scenarios) . . . . .  | 5         |
| 4.2      | DIM Design Comparison by MSE . . . . .         | 6         |
| 4.3      | DIM: MSE by Design and Neighbor Type . . . . . | 7         |
| 4.4      | DIM: MSE Heatmap (Rho x Design) . . . . .      | 8         |
| 4.5      | DIM: Design Rank Frequency . . . . .           | 8         |
| 4.6      | DIM: Robustness Profile . . . . .              | 8         |
| 4.7      | DIM: Coverage by Design . . . . .              | 9         |
| 4.8      | DIM: Bias-Variance Decomposition . . . . .     | 10        |
| <b>5</b> | <b>Results: Maximum Likelihood (MLE)</b>       | <b>10</b> |
| 5.1      | Overall Summary (All MLE Scenarios) . . . . .  | 10        |
| 5.2      | MLE Design Comparison by MSE . . . . .         | 11        |
| 5.3      | MLE: MSE by Design and Neighbor Type . . . . . | 12        |
| 5.4      | MLE: MSE Heatmap . . . . .                     | 13        |
| 5.5      | MLE: Design Rank Frequency . . . . .           | 13        |
| 5.6      | MLE: Robustness Profile . . . . .              | 13        |
| 5.7      | MLE: Coverage by Design . . . . .              | 14        |
| <b>6</b> | <b>DIM vs. MLE Comparison</b>                  | <b>15</b> |
| 6.1      | Side-by-Side Metrics . . . . .                 | 15        |
| 6.2      | Coverage Comparison by Estimator . . . . .     | 16        |
| 6.3      | Key Observations . . . . .                     | 16        |

|           |                                                                              |           |
|-----------|------------------------------------------------------------------------------|-----------|
| <b>7</b>  | <b>Sensitivity Analysis</b>                                                  | <b>17</b> |
| 7.1       | Parameter Sensitivity (DIM)                                                  | 17        |
| 7.2       | Effect of Rho on Design Rankings (DIM)                                       | 17        |
| 7.3       | Incidence Mode Comparison                                                    | 29        |
| <b>8</b>  | <b>Subgroup Results by Incidence Mode</b>                                    | <b>30</b> |
| 8.1       | DIM: iid Uniform                                                             | 30        |
| 8.2       | DIM: Poisson ( $\rho_X = 0.20$ )                                             | 31        |
| 8.3       | DIM: Poisson ( $\rho_X = 0.50$ )                                             | 31        |
| 8.4       | DIM: Spatial ( $\rho_X = 0.20$ )                                             | 31        |
| 8.5       | DIM: Spatial ( $\rho_X = 0.50$ )                                             | 31        |
| <b>9</b>  | <b>Tau Sensitivity Analysis</b>                                              | <b>32</b> |
| <b>10</b> | <b>Conclusions</b>                                                           | <b>35</b> |
| 10.1      | Summary of Findings                                                          | 35        |
| 10.2      | Limitations and Planned Extensions                                           | 36        |
| <b>11</b> | <b>References</b>                                                            | <b>36</b> |
| <b>12</b> | <b>Appendix: Result File Locations</b>                                       | <b>36</b> |
| ##        | Loading results from: sim_results_DIM_combined_20260304_195321.rds           |           |
| ##        | Loading results from: sim_results_MLE_tau_sweep_combined_20260924_025509.rds |           |

# 1 Introduction

This report summarizes a simulation study evaluating **eight treatment assignment designs** for Spatial Cluster Randomized Trials (CRTs) when outcome incidence is heterogeneous and spatially structured. The central research question is:

*Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment strategy minimizes estimation error for the direct intervention effect  $\hat{\tau}$  across a range of spatial dependence and spillover configurations?*

**Application context:** Sudden Unexpected Death (SUD) in North Carolina counties. The Poisson incidence mode uses an SUD base rate of 35 per 100,000 (Mirzaei et al.), consistent with county-level analyses by Gan et al. and Watson et al.

The primary estimator is Maximum Likelihood Estimation via a spatial autoregressive model (MLE), fit to the 2026-09 revised simulation. Difference in Means (DIM) appears only as a naive baseline, and its results are **pre-revision** (March 2026 run: old data-generating process, 6 designs), so DIM and MLE numbers are not like-for-like. Queen contiguity is the primary neighbor structure; rook is a sensitivity analysis. The primary MLE results use the  $\tau = 1.0$  scenario (2,560 parameter scenarios); a tau sensitivity analysis across  $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$  (12,800 total scenarios) is presented in Section 9.

## 2 Mathematical Framework

### 2.1 Study Region and Spatial Structure

The simulation operates on a **regular lattice grid**  $\mathcal{G} = \{s_1, \dots, s_N\}$  on  $[1, G] \times [1, G]$ , with  $N = G^2$  clusters ( $G = 10$ ,  $N = 100$ ).

Two contiguity structures define neighbor relationships:

- **Rook contiguity:**  $s_j \in \mathcal{N}(s_i)$  iff  $s_i$  and  $s_j$  share an edge (up to 4 neighbors)
- **Queen contiguity:**  $s_j \in \mathcal{N}(s_i)$  iff  $s_i$  and  $s_j$  share an edge or corner (up to 8 neighbors)

The spatial weight matrix  $\mathbf{W}$  has entries  $w_{ij} = |\mathcal{N}(s_i)|^{-1}$  if  $s_j \in \mathcal{N}(s_i)$  and 0 otherwise (row-standardized).

### 2.2 Outcome Data Generating Process

Outcomes follow a **Spatial Durbin Model (SDM)**:

$$\mathbf{Y} = (\mathbf{I} - \rho \mathbf{W})^{-1} [\tau \mathbf{Z} + \gamma \cdot \text{Spill}(\mathbf{Z}) + \beta \mathbf{X} + \varepsilon]$$

**Parameter definitions:**

| Symbol   | Role                               | Value / Range                                |
|----------|------------------------------------|----------------------------------------------|
| $\tau$   | Direct treatment effect (estimand) | $\{0.8, 1.0, 1.5, 2.0, 3.0\}$ ; primary: 1.0 |
| $\rho$   | Outcome spatial autocorrelation    | $\{0.00, 0.01, 0.20, 0.50\}$                 |
| $\gamma$ | Spillover magnitude                | $\{0.5, 0.6, 0.7, 0.8\}$                     |
| $\beta$  | Incidence effect on outcome        | 1.0 (fixed)                                  |
| $\sigma$ | Residual standard deviation        | 1.0 (fixed)                                  |

### Spillover construction:

$$\text{Spill}_i(\mathbf{Z}) = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|}$$

Two spillover modes are evaluated: - **control\_only**:  $\gamma$  applied only to units with  $Z_i = 0$  - **both**:  $\gamma$  applied to all units regardless of treatment status

Stationarity requires  $|\rho| < 1/\lambda_{\max}(\mathbf{W})$ ; all swept  $\rho$  values satisfy this condition.

## 2.3 Incidence Generation

Three modes generate the cluster-level covariate  $X_i \in [0, 1]$ , each representing a different assumption about how historical incidence arises:

### Mode 1 — iid Uniform:

$$X_i \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$$

No spatial structure; serves as the baseline. One config ( $\rho_X = N/A$ ).

### Mode 2 — Spatial (SAR filter):

$$\mathbf{X}^* = (\mathbf{I} - \rho_X \mathbf{W})^{-1} \mathbf{u}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$X_i = \Phi(X_i^*) \in (0, 1)$$

Spatially correlated via SAR filter; probability integral transform ( $\Phi$ ) ensures  $X_i$  is marginally  $\text{Uniform}(0, 1)$  but spatially dependent. Two configs:  $\rho_X \in \{0.20, 0.50\}$ .

### Mode 3 — Poisson (count rates):

1. Spatially correlated log-rates:  $\log \lambda_i^* = \mu_0 + [(\mathbf{I} - \rho_X \mathbf{W})^{-1} \mathbf{u}]_i$
2. Poisson counts:  $C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$  where  $P_i = 100,000$  (equal population per cluster;  $\lambda_i^* P_i \approx 35$  expected deaths)
3. Observed rates:  $R_i = C_i / P_i$
4. Rank-normalized:  $X_i = (\text{rank}(R_i) - 0.5) / N$

Base rate  $\mu_0 = \log(35/100,000)$  mimics SUD incidence. Two configs:  $\rho_X \in \{0.20, 0.50\}$ .

**Critical design invariant:** Incidence surfaces  $\mathbf{X}_1, \dots, \mathbf{X}_{10}$  are generated **once per (mode,  $\rho_X$ ) configuration** and shared by all ( $\rho$ ,  $\gamma$ , nb\_type, spillover type) scenarios of that configuration; spatial model parameters affect the outcome DGP, not the incidence itself. **Matched surfaces:** for each surface  $k$ , treatment is assigned from  $\mathbf{X}_k$ , and the outcome and the analysis use the same  $\mathbf{X}_k$ . Reported metrics are therefore averages over incidence surfaces.

## 2.4 Treatment Assignment Designs

Eight assignment strategies are evaluated, representing qualitatively distinct approaches to allocating clusters to treatment (all treat exactly 50 of 100 except Isolation Buffer):

| ID | Name                 | Deterministic? | Key Feature                                                               |
|----|----------------------|----------------|---------------------------------------------------------------------------|
| 1  | Checkerboard         | Yes            | Alternating grid; maximizes spatial separation                            |
| 2  | High Incidence Focus | No             | Treats the top-50 incidence clusters (random tie-breaking)                |
| 3  | Saturation Quadrants | No             | Random permutation of saturations $\{0.2, 0.4, 0.6, 0.8\}$ over quadrants |

| ID | Name                                  | Deterministic? | Key Feature                                                                                               |
|----|---------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| 4  | Isolation Buffer                      | No             | Greedy random maximal independent set; no adjacent treated clusters (mean treated: rook 38.2, queen 21.4) |
| 5  | 2x2 Blocking                          | No             | 2 of 4 treated within each 2x2 spatial block                                                              |
| 6  | Balanced Quartiles                    | No             | Stratified by incidence quartile; exactly 50 treated                                                      |
| 7  | Balanced Halves                       | No             | Median split; 25 treated per half                                                                         |
| 8  | Incidence-Guided Saturation Quadrants | No             | Quadrant saturations {0.8, 0.6, 0.4, 0.2} by rank of quadrant mean incidence                              |

Ties in incidence are broken at random, freshly for every design draw. Checkerboard is the only **deterministic** design: its assignment is identical on every draw, but each fit still receives its own outcome noise.

## 2.5 Estimation Methods

**Difference in Means (DIM):**

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1} \sum_{Z_i=1} Y_i - \frac{1}{n_0} \sum_{Z_i=0} Y_i$$

95% CI via Neyman variance estimator:

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$$

**Maximum Likelihood (MLE via SAR):**

Fits the spatial lag model  $Y \sim Z + \text{Spill} + X$  by maximum likelihood with `fit_sar_lag()` (`04_estimation.R`), which matches `spatialreg::lagsarlm(method = "eigen")` to about  $10^{-7}$ , simultaneously estimating  $\rho$ ,  $\tau$ , and regression coefficients. The true spillover covariate  $\text{Spill}(\mathbf{Z})$  is included as an **oracle regressor** (i.e., the researcher is assumed to know the true spillover structure). 95% Wald CIs use the ML standard error. A non-oracle fit ( $Y \sim Z + X$ ) was run as a sensitivity analysis; it is not shown in this report. Under rook contiguity, Checkerboard has  $W\mathbf{Z} = \mathbf{1} - \mathbf{Z}$ , so the oracle Spill column is aliased and  $\tau$  is not identified; those estimates are kept but flagged.

## 3 Simulation Algorithm

### 3.1 Parameter Grid

Table 1: Simulation parameter grid. Primary tau=1.0 slice: 2,560 scenarios. Full tau-sweep (tau in 0.8, 1.0, 1.5, 2.0, 3.0): 12,800 total scenarios.

| Parameter                           | Values                                   | Levels                                    |
|-------------------------------------|------------------------------------------|-------------------------------------------|
| Incidence mode                      | iid, spatial, poisson                    | 5 configs (1 iid + 2 spatial + 2 poisson) |
| rho_X (incidence spatial autocorr.) | N/A (iid); 0.20, 0.50 (spatial, poisson) | —                                         |
| nb_type (contiguity)                | rook, queen                              | 2                                         |
| rho (outcome spatial autocorr.)     | 0.00, 0.01, 0.20, 0.50                   | 4                                         |
| gamma (spillover magnitude)         | 0.5, 0.6, 0.7, 0.8                       | 4                                         |
| spill_type                          | control_only, both                       | 2                                         |

|              |                        |                                                |
|--------------|------------------------|------------------------------------------------|
| design_id    | 1, 2, 3, 4, 5, 6, 7, 8 | 8                                              |
| <b>TOTAL</b> |                        | <b>2,560 scenarios (primary tau=1.0 slice)</b> |

### 3.2 Resampling Structure

For each MLE scenario, the simulation uses  $K = 10$  incidence surfaces and  $J = 25$  design draws per surface; every one of the  $K \cdot J = 250$  fits has its own outcome noise column. The DIM baseline (pre-revision) used a different scheme:

| Mode               | Design draws   | Outcome draws | Fits / scenario |
|--------------------|----------------|---------------|-----------------|
| DIM (pre-revision) | 25             | 100           | 2,500           |
| MLE                | 25 per surface | 10 surfaces   | 250             |

### 3.3 Evaluation Metrics

For each scenario, across the  $n$  valid fits  $i = 1, \dots, n$ :

$$\text{Bias} = \bar{\hat{\tau}} - \tau, \quad \text{SD} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\hat{\tau}_i - \bar{\hat{\tau}})^2}$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i - \tau)^2$$

$$\text{Coverage} = \frac{1}{n} \sum_{i=1}^n \mathbf{1} [\tau \in (\hat{\tau}_{i,L}, \hat{\tau}_{i,U})]$$

Monte Carlo standard errors are computed from the 10 surface-level means (the surface is the primary sampling unit), not from the 250 fits treated as independent.

### 3.4 Per-Draw Reproducibility

Every random draw is preceded by a key-based seed, `set.seed(digest::digest2int(key))`, where the key joins the fields that define the draw with “|”: (“X”, mode,  $\rho_X$ ,  $k$ ) for surface  $k$ ; (“Z”, mode,  $\rho_X$ , nb\_type,  $\rho$ ,  $\gamma$ , spill\_type,  $k$ ,  $d$ ) for design draws; (“eps”, mode,  $\rho_X$ , nb\_type,  $\rho$ ,  $\gamma$ , spill\_type) for outcome noise.

Each draw depends only on its key, so results are identical in any run order. Within a block, all designs share the same noise and incidence surfaces, and every  $\tau$  level reuses the same assignments and noise (common random numbers).

## 4 Results: Difference in Means (DIM)

DIM is a naive baseline. All results in this section are **pre-revision** (March 2026 run: old data-generating process, 6 designs, no Balanced Halves or Incidence-Guided Saturation Quadrants); design labels use the current numbering. Tables pool rook and queen.

### 4.1 Overall Summary (All DIM Scenarios)

Table 2: DIM (pre-revision): Summary statistics across all 1920 scenarios

| Metric   | Min    | Mean   | Max   |
|----------|--------|--------|-------|
| Bias     | -0.883 | -0.204 | 0.558 |
| MSE      | 0.032  | 0.152  | 0.799 |
| Coverage | 0.000  | 0.720  | 0.990 |

## 4.2 DIM Design Comparison by MSE

Table 3: DIM (pre-revision): Design performance averaged across all scenarios and incidence modes

| Design          | Mean Bias     | Median MSE   | Mean MSE     | Mean Coverage |
|-----------------|---------------|--------------|--------------|---------------|
| <b>Design 6</b> | <b>-0.128</b> | <b>0.091</b> | <b>0.095</b> | <b>0.821</b>  |
| Design 2        | -0.076        | 0.081        | 0.101        | 0.822         |
| Design 3        | -0.011        | 0.090        | 0.111        | 0.802         |
| Design 5        | -0.253        | 0.101        | 0.128        | 0.736         |
| Design 4        | -0.310        | 0.133        | 0.163        | 0.702         |
| Design 1        | -0.448        | 0.253        | 0.314        | 0.446         |

### 4.3 DIM: MSE by Design and Neighbor Type

## Distribution of Estimation Error (MSE) by Spatial Contiguity

Incidence: iid Uniform

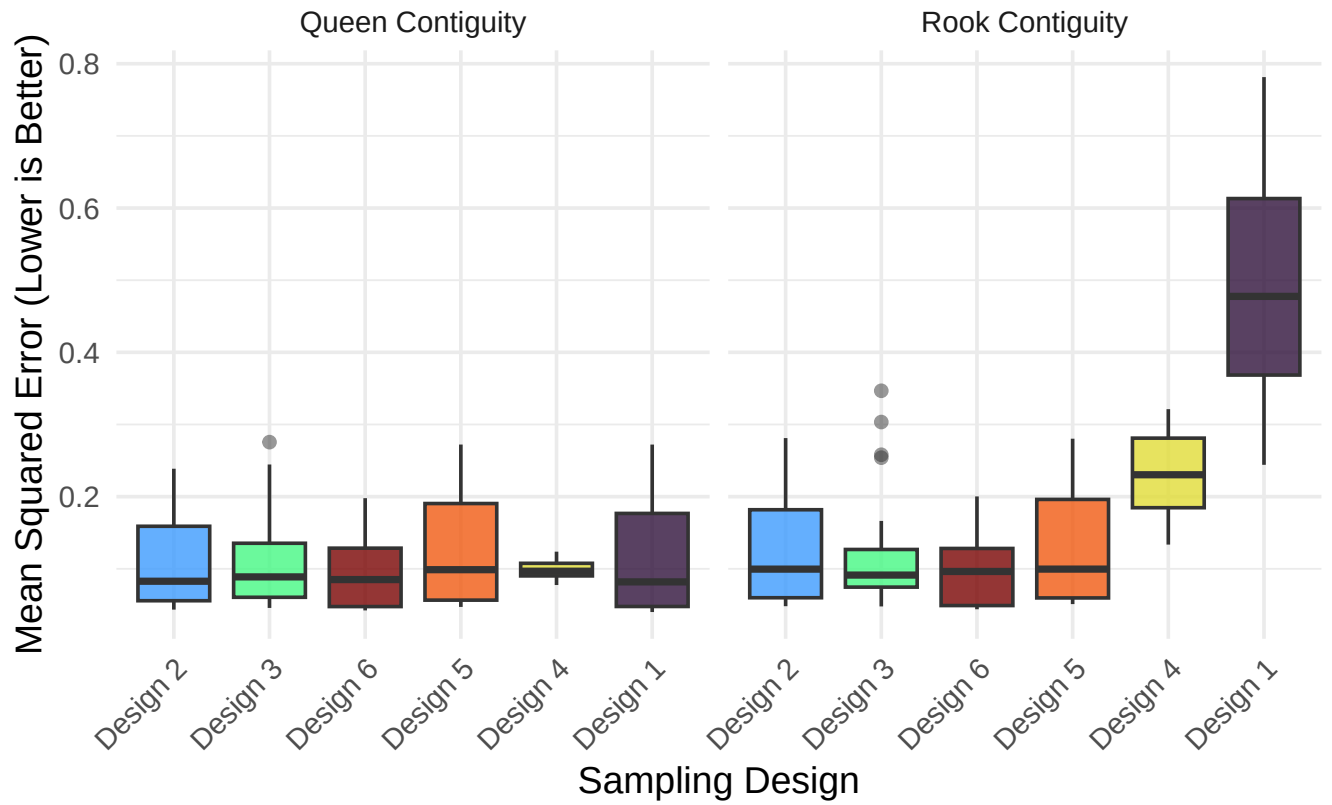


Figure 1: DIM (pre-revision): Distribution of MSE by design, faceted by spatial contiguity structure. Boxes sorted by median MSE within each panel.

#### 4.4 DIM: MSE Heatmap (Rho x Design)

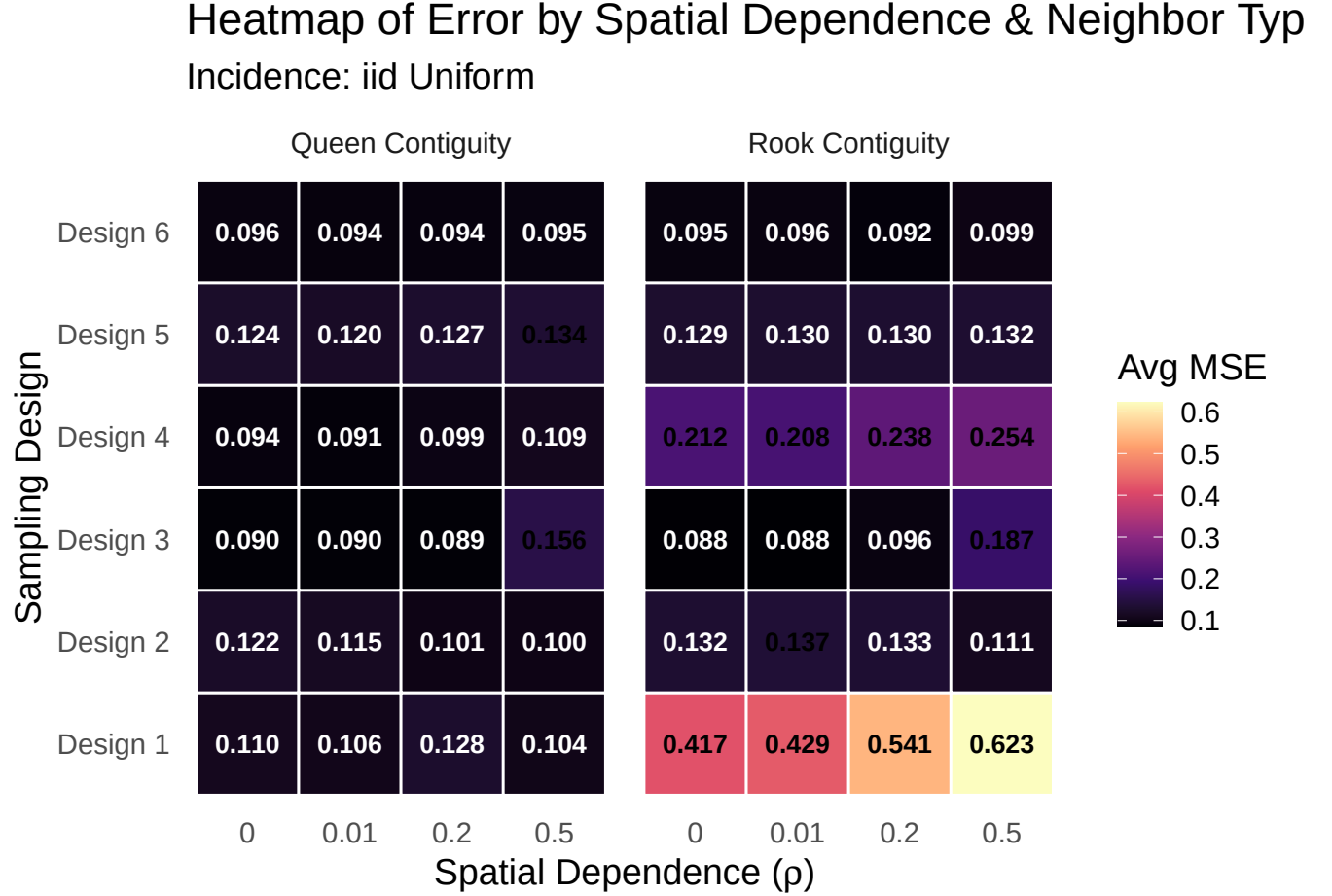


Figure 2: DIM (pre-revision) MSE heatmap for iid Uniform incidence. Cells show average MSE across spillover type and gamma, faceted by neighbor type.

#### 4.5 DIM: Design Rank Frequency

Table 4: DIM (pre-revision): Design rank frequency (%) for iid Uniform incidence. Each cell = % of scenarios where design achieved that rank (Rank 1 = lowest MSE).

| Design   | Rank_1 | Rank_2 | Rank_3 | Rank_4 | Rank_5 | Rank_6 |
|----------|--------|--------|--------|--------|--------|--------|
| Design 3 | 32.8   | 29.7   | 7.8    | 9.4    | 14.1   | 6.2    |
| Design 6 | 28.1   | 34.4   | 31.2   | 6.2    | 0.0    | 0.0    |
| Design 4 | 15.6   | 7.8    | 4.7    | 4.7    | 48.4   | 18.8   |
| Design 2 | 12.5   | 4.7    | 21.9   | 39.1   | 20.3   | 1.6    |
| Design 1 | 10.9   | 7.8    | 4.7    | 14.1   | 6.2    | 56.2   |
| Design 5 | 0.0    | 15.6   | 29.7   | 26.6   | 10.9   | 17.2   |

#### 4.6 DIM: Robustness Profile

Table 5: DIM (pre-revision): Robustness profile for iid Uniform incidence. Sorted by Median MSE. IQR = interquartile range across scenarios.

| Design   | Best_MSE | Q25_MSE | Median_MSE | Q75_MSE | Worst_MSE | IQR_MSE |
|----------|----------|---------|------------|---------|-----------|---------|
| Design 2 | 0.0434   | 0.0574  | 0.0897     | 0.1754  | 0.2812    | 0.1180  |
| Design 3 | 0.0458   | 0.0605  | 0.0904     | 0.1306  | 0.3467    | 0.0701  |
| Design 6 | 0.0423   | 0.0478  | 0.0945     | 0.1284  | 0.2003    | 0.0806  |
| Design 5 | 0.0472   | 0.0575  | 0.0996     | 0.1933  | 0.2803    | 0.1358  |
| Design 4 | 0.0775   | 0.0969  | 0.1287     | 0.2295  | 0.3213    | 0.1326  |
| Design 1 | 0.0401   | 0.0962  | 0.2441     | 0.4764  | 0.7815    | 0.3801  |

#### 4.7 DIM: Coverage by Design

### 95% CI Coverage Probability by Design

Incidence: iid Uniform | Red dashed = nominal 95%

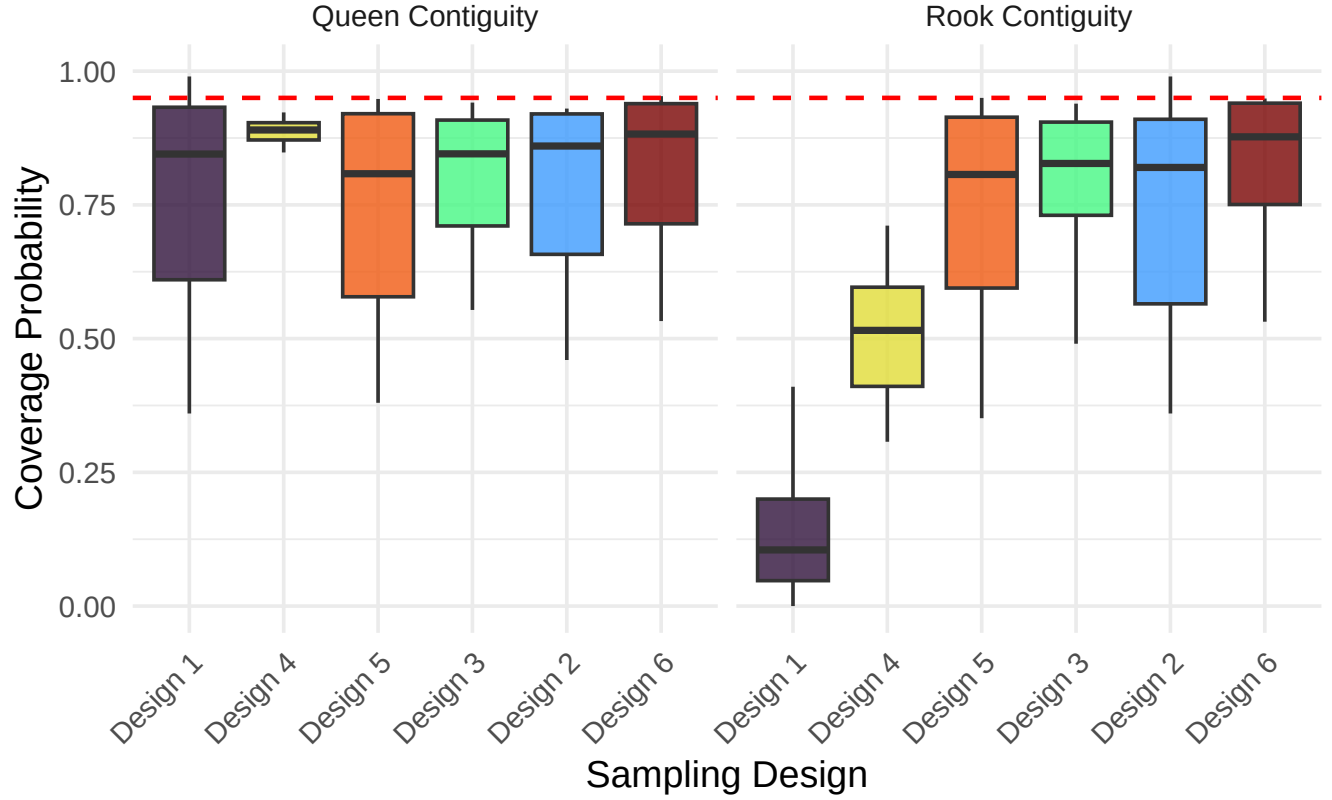


Figure 3: DIM (pre-revision): Coverage probability distribution by design across all scenarios. Red dashed line = nominal 95% target.

## 4.8 DIM: Bias-Variance Decomposition

### Bias-Variance Decomposition of MSE

Incidence: iid Uniform

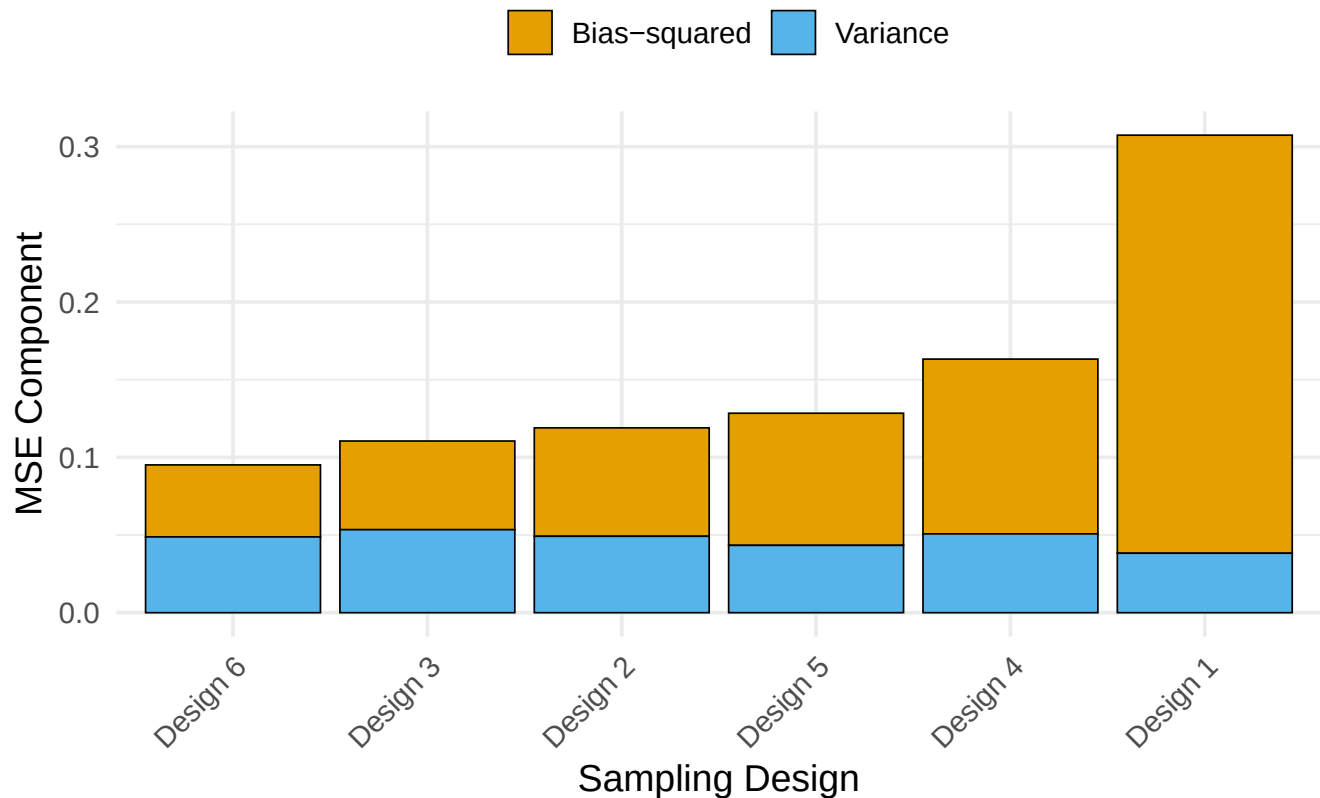


Figure 4: DIM (pre-revision): Bias-squared (dark) and variance (light) components of MSE per design for iid Uniform incidence. Total bar height = MSE.

---

## 5 Results: Maximum Likelihood (MLE)

### 5.1 Overall Summary (All MLE Scenarios)

Table 6: MLE: Summary statistics across all 2560  $\tau = 1.0$  scenarios, pooled over rook and queen and over incidence modes. The minimum coverage comes from Checkerboard under rook, where  $\tau$  is not identified.

| Metric   | Min    | Mean   | Max   |
|----------|--------|--------|-------|
| Bias     | -0.827 | -0.028 | 0.260 |
| MSE      | 0.032  | 0.219  | 2.843 |
| Coverage | 0.000  | 0.890  | 0.990 |

|           |    |       |    |
|-----------|----|-------|----|
| Fail Rate | NA | 0.000 | NA |
|-----------|----|-------|----|

## 5.2 MLE Design Comparison by MSE

Table 7: MLE: Design performance at  $\tau = 1.0$  by neighbor type (queen primary, rook sensitivity), averaged across incidence modes. Checkerboard under rook:  $\tau$  not identified ( $WZ = 1 - Z$ ); estimates kept but flagged.

| Neighbors    | Design          | Mean Bias    | Median MSE   | Mean MSE     | Mean Coverage | Fail Rate |
|--------------|-----------------|--------------|--------------|--------------|---------------|-----------|
| <b>queen</b> | <b>Design 3</b> | <b>0.025</b> | <b>0.079</b> | <b>0.090</b> | <b>0.936</b>  | <b>0</b>  |
| queen        | Design 8        | 0.032        | 0.080        | 0.091        | 0.937         | 0         |
| queen        | Design 7        | 0.018        | 0.114        | 0.130        | 0.936         | 0         |
| queen        | Design 6        | 0.017        | 0.115        | 0.131        | 0.936         | 0         |
| queen        | Design 4        | 0.003        | 0.159        | 0.160        | 0.939         | 0         |
| queen        | Design 2        | 0.031        | 0.234        | 0.245        | 0.938         | 0         |
| queen        | Design 5        | 0.010        | 0.271        | 0.319        | 0.939         | 0         |
| queen        | Design 1        | 0.001        | 0.899        | 1.087        | 0.941         | 0         |
| rook         | Design 3        | 0.011        | 0.065        | 0.071        | 0.940         | 0         |
| rook         | Design 8        | 0.018        | 0.069        | 0.072        | 0.942         | 0         |
| rook         | Design 6        | 0.008        | 0.078        | 0.084        | 0.940         | 0         |
| rook         | Design 7        | 0.008        | 0.079        | 0.084        | 0.940         | 0         |
| rook         | Design 5        | 0.004        | 0.117        | 0.128        | 0.938         | 0         |
| rook         | Design 4        | 0.003        | 0.133        | 0.133        | 0.939         | 0         |
| rook         | Design 2        | 0.018        | 0.194        | 0.206        | 0.939         | 0         |
| rook         | Design 1        | -0.652       | 0.470        | 0.480        | 0.150         | 0         |

### 5.3 MLE: MSE by Design and Neighbor Type

#### Distribution of Estimation Error (MSE) by Spatial Contiguity

Incidence: iid Uniform

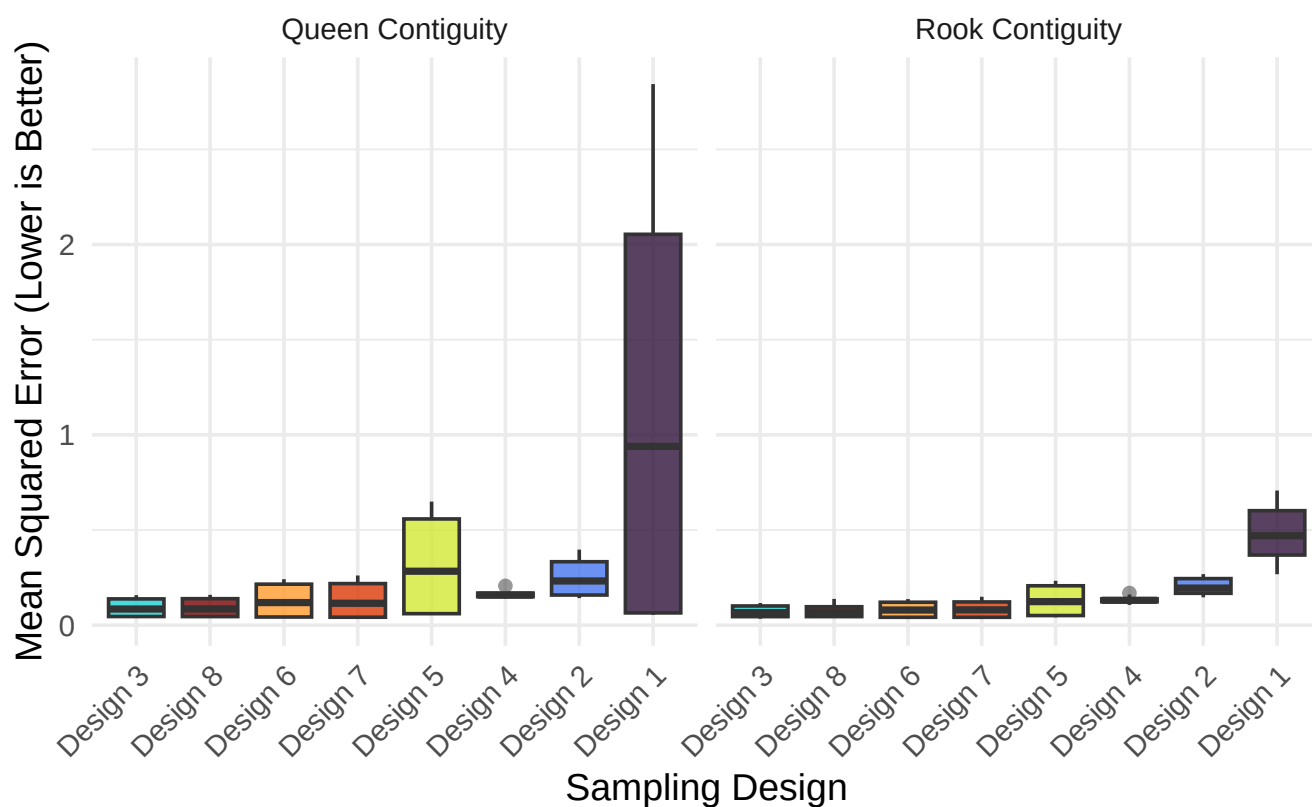


Figure 5: MLE: Distribution of MSE by design at  $\tau = 1.0$  (iid Uniform incidence), faceted by spatial contiguity (queen primary, rook sensitivity). Checkerboard under rook:  $\tau$  not identified; estimates kept but flagged.

## 5.4 MLE: MSE Heatmap

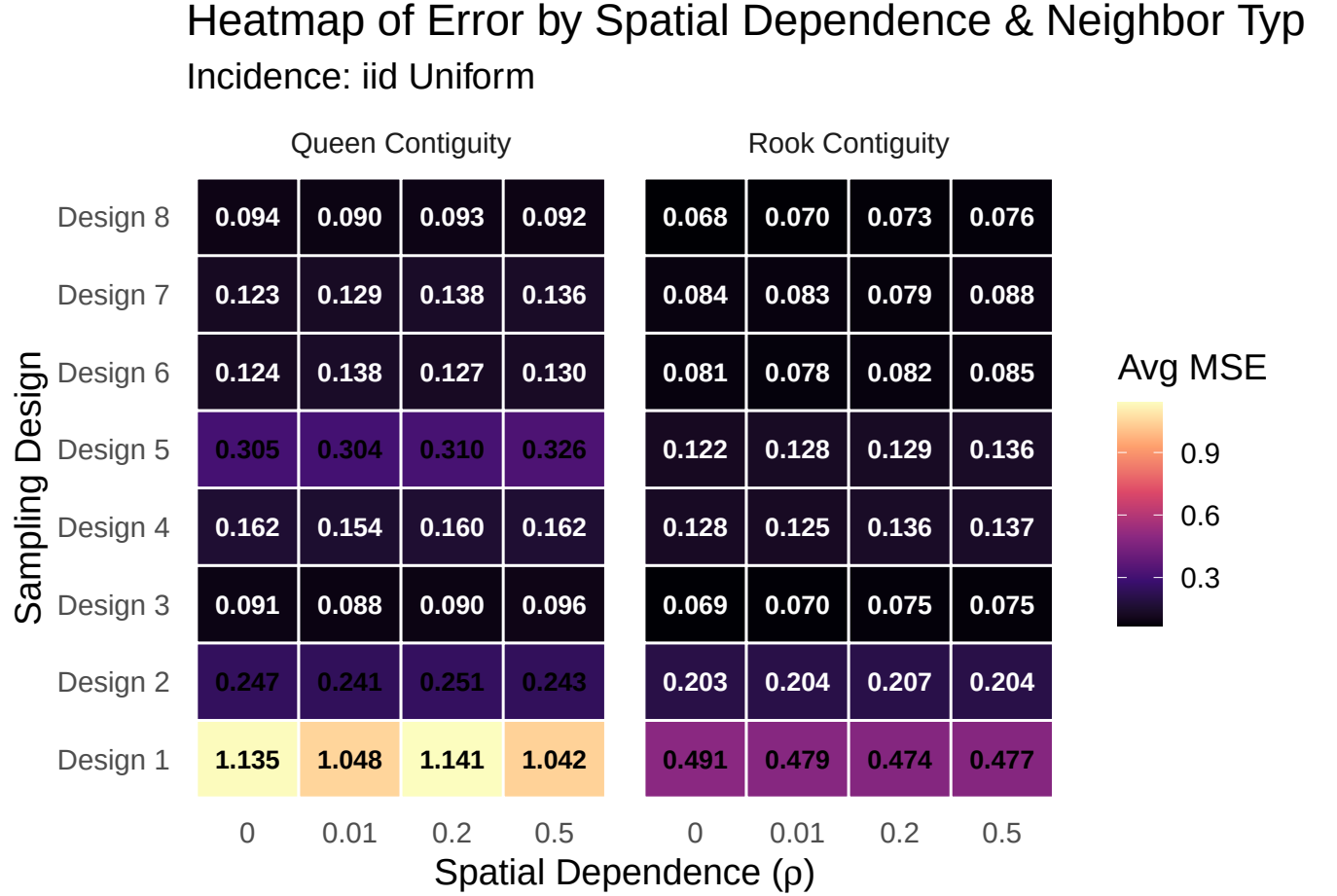


Figure 6: MLE MSE heatmap for iid Uniform incidence at  $\tau = 1.0$ , faceted by neighbor type. Cells show average MSE across spillover type and gamma.

## 5.5 MLE: Design Rank Frequency

Table 8: MLE: Design rank frequency (%) for iid Uniform incidence, queen contiguity,  $\tau = 1.0$ .

| Design   | Rank_5 | Rank_6 | Rank_8 | Rank_7 | Rank_1 | Rank_2 | Rank_3 | Rank_4 |
|----------|--------|--------|--------|--------|--------|--------|--------|--------|
| Design 3 | 0.0    | 0.0    | 0      | 0      | 40.6   | 25.0   | 18.8   | 15.6   |
| Design 8 | 0.0    | 0.0    | 0      | 0      | 21.9   | 31.2   | 25.0   | 21.9   |
| Design 7 | 25.0   | 0.0    | 0      | 0      | 18.8   | 15.6   | 12.5   | 28.1   |
| Design 6 | 25.0   | 0.0    | 0      | 0      | 12.5   | 25.0   | 3.1    | 34.4   |
| Design 4 | 0.0    | 0.0    | 25     | 25     | 6.2    | 3.1    | 40.6   | 0.0    |
| Design 1 | 21.9   | 28.1   | 50     | 0      | 0.0    | 0.0    | 0.0    | 0.0    |
| Design 2 | 0.0    | 50.0   | 25     | 25     | 0.0    | 0.0    | 0.0    | 0.0    |
| Design 5 | 28.1   | 21.9   | 0      | 50     | 0.0    | 0.0    | 0.0    | 0.0    |

## 5.6 MLE: Robustness Profile

Table 9: MLE: Robustness profile for iid Uniform incidence, queen contiguity,  $\tau = 1.0$ .

| Design   | Best_MSE | Q25_MSE | Median_MSE | Q75_MSE | Worst_MSE | IQR_MSE |
|----------|----------|---------|------------|---------|-----------|---------|
| Design 3 | 0.0377   | 0.0454  | 0.0848     | 0.1381  | 0.1576    | 0.0927  |
| Design 8 | 0.0392   | 0.0451  | 0.0853     | 0.1392  | 0.1594    | 0.0941  |
| Design 7 | 0.0373   | 0.0417  | 0.1150     | 0.2185  | 0.2610    | 0.1767  |
| Design 6 | 0.0381   | 0.0428  | 0.1187     | 0.2153  | 0.2416    | 0.1725  |
| Design 4 | 0.1410   | 0.1495  | 0.1580     | 0.1686  | 0.2065    | 0.0191  |
| Design 2 | 0.1423   | 0.1584  | 0.2323     | 0.3334  | 0.3970    | 0.1750  |
| Design 5 | 0.0544   | 0.0604  | 0.2835     | 0.5581  | 0.6493    | 0.4977  |
| Design 1 | 0.0547   | 0.0645  | 0.9394     | 2.0536  | 2.8430    | 1.9892  |

## 5.7 MLE: Coverage by Design

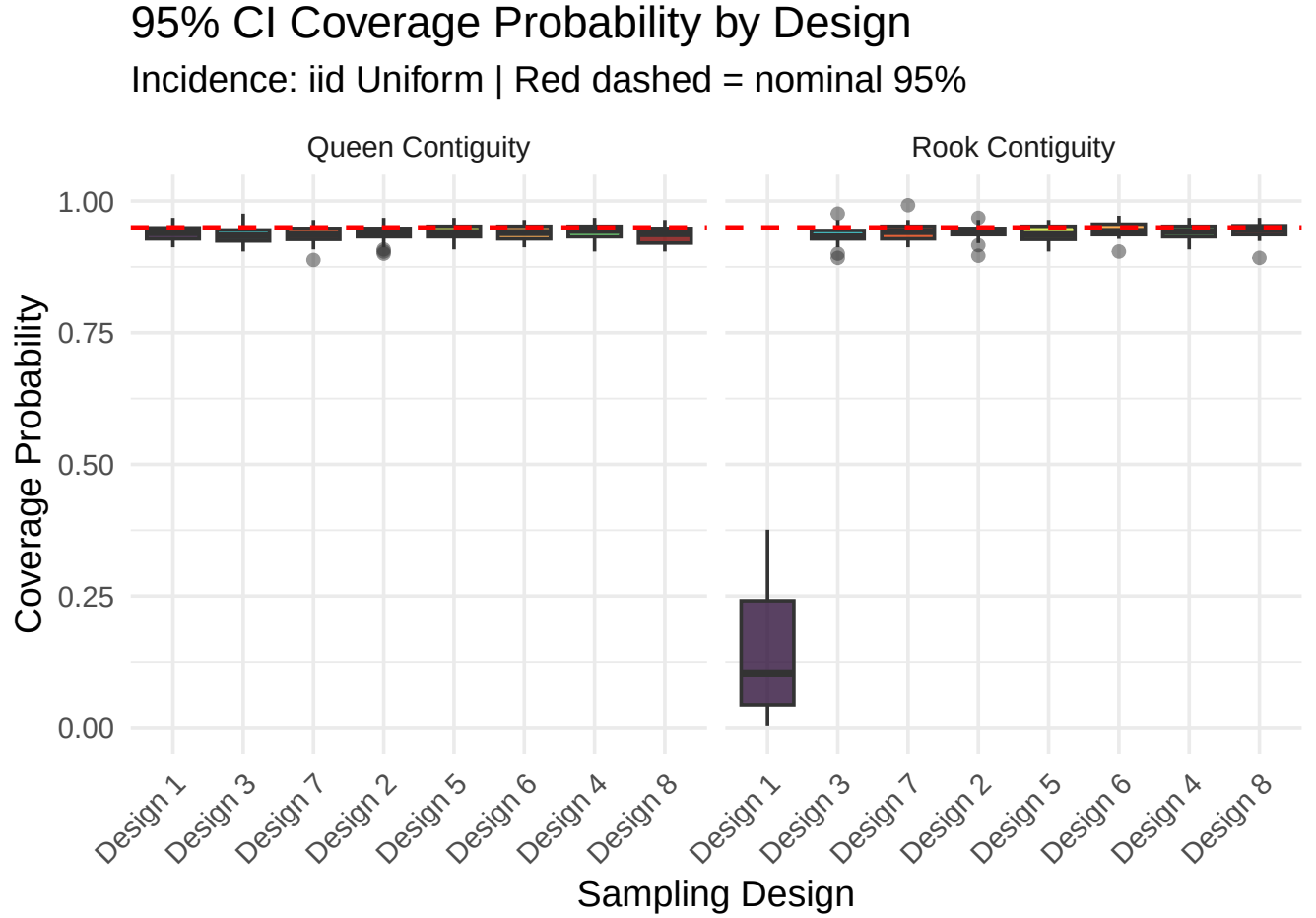


Figure 7: MLE: Coverage probability distribution by design (iid Uniform incidence,  $\tau = 1.0$ ), faceted by neighbor type. Coverage is near-nominal for every design except Checkerboard under rook, where  $\tau$  is not identified. Red dashed line = 95%.

## 6 DIM vs. MLE Comparison

### 6.1 Side-by-Side Metrics

Table 10: DIM vs. MLE: Average MSE and coverage by design, pooled over rook and queen and over incidence modes (MLE at  $\tau = 1.0$ ). DIM is the pre-revision baseline (old data-generating process, 6 designs), so the two columns are not like-for-like; Design 7 (Balanced Halves) and Design 8 (Incidence-Guided Saturation Quadrants) were not in the DIM run and are NA by construction. Pooled MLE values for Checkerboard mix queen with rook, where  $\tau$  is not identified.

| Design   | DIM (pre-revision) |              | MLE     |              |
|----------|--------------------|--------------|---------|--------------|
|          | DIM MSE            | DIM Coverage | MLE MSE | MLE Coverage |
| Design 3 | 0.111              | 0.802        | 0.080   | 0.938        |
| Design 8 | NA                 | NA           | 0.081   | 0.939        |
| Design 6 | 0.095              | 0.821        | 0.107   | 0.938        |
| Design 7 | NA                 | NA           | 0.107   | 0.938        |
| Design 4 | 0.163              | 0.702        | 0.147   | 0.939        |
| Design 5 | 0.128              | 0.736        | 0.224   | 0.938        |
| Design 2 | 0.101              | 0.822        | 0.226   | 0.938        |
| Design 1 | 0.314              | 0.446        | 0.783   | 0.545        |

## 6.2 Coverage Comparison by Estimator

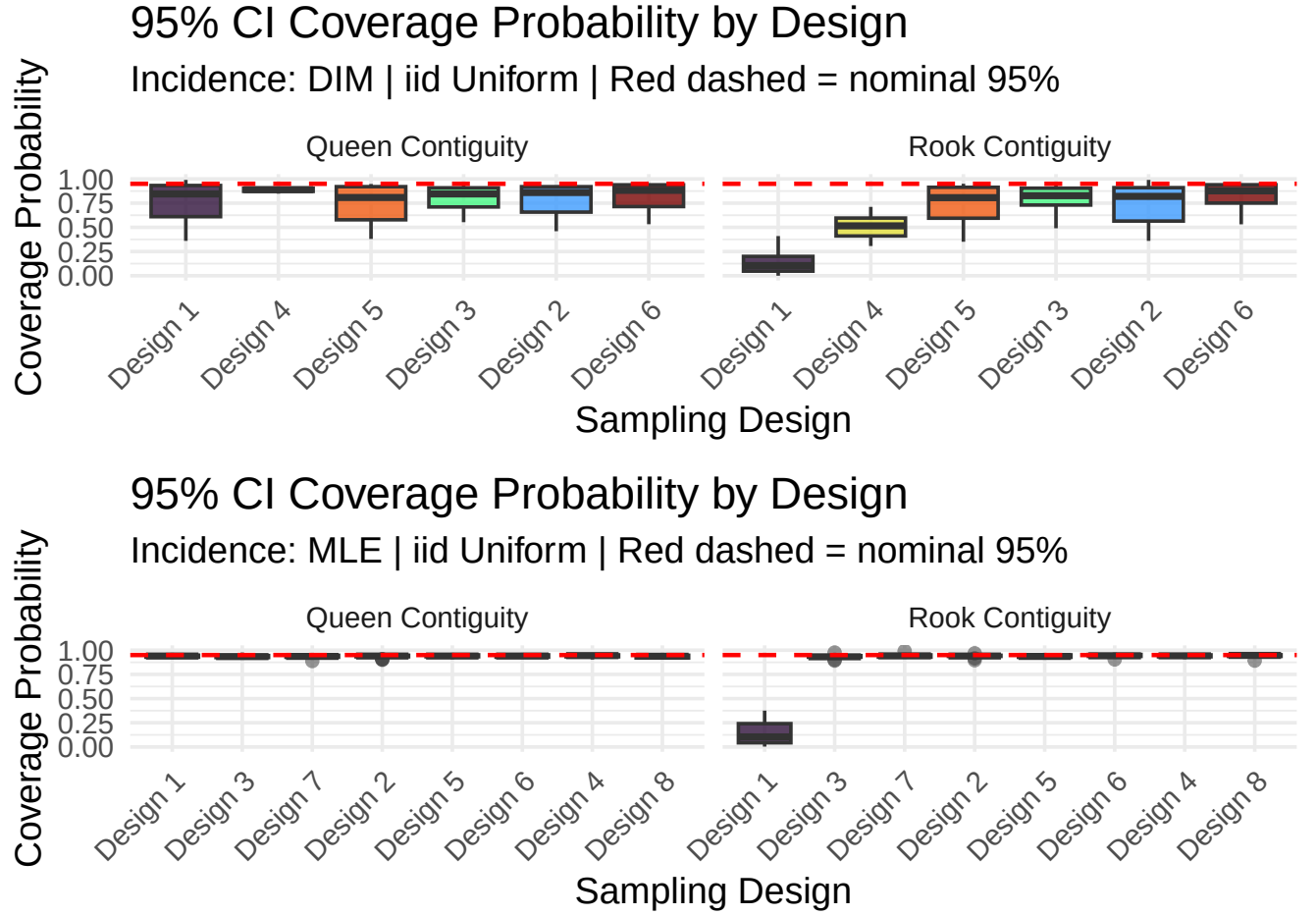


Figure 8: Coverage comparison: DIM (top, pre-revision baseline) vs MLE (bottom,  $\tau = 1.0$ ) for the iid Uniform incidence configuration. MLE achieves near-nominal coverage for every design except Checkerboard under rook; DIM coverage is mostly below 0.95.

## 6.3 Key Observations

- **Coverage does not separate the designs.** MLE coverage averages 0.936–0.942 for every design and neighbor type except Checkerboard under rook, i.e. near-nominal (a finite-sample Wald shortfall of about one point). MSE is what separates the designs.
- **DIM baseline (pre-revision):** mean coverage 0.72 across its 1920 scenarios. It was run under the old data-generating process, so it is not a like-for-like comparison with the MLE results.
- **Checkerboard (Design 1):** highest MSE under queen (1.087 at  $\tau = 1$ ), with mean bias 0.001, so almost all of it is variance: the oracle model separates  $\tau$  from the spillover term only weakly under a checkerboard. Under rook,  $WZ = \mathbf{1} - Z$ , so  $\tau$  is not identified: bias -0.652 (approximately  $-\gamma$ ), coverage 0.150. These estimates are kept but flagged.
- **Lowest MSE (queen):** Saturation Quadrants (0.090) and Incidence-Guided Saturation Quadrants (0.091), which are statistically indistinguishable (eight-design comparisons in [results/eight\\_design\\_supplementary/](#))

## 7 Sensitivity Analysis

### 7.1 Parameter Sensitivity (DIM)

Table 11: DIM (pre-revision): Eta-squared (proportion of MSE variance explained) by parameter, for iid Uniform incidence. Higher values indicate parameters where design choice matters more.

| Design   | Rho    | Gamma  | Spillover_Type | Neighbor_Type |
|----------|--------|--------|----------------|---------------|
| Design 1 | 0.0350 | 0.1088 | 0.0202         | 0.7220        |
| Design 2 | 0.0147 | 0.1018 | 0.7598         | 0.0172        |
| Design 3 | 0.2809 | 0.0582 | 0.0005         | 0.0042        |
| Design 4 | 0.0274 | 0.1661 | 0.0001         | 0.7196        |
| Design 5 | 0.0015 | 0.1172 | 0.7968         | 0.0006        |
| Design 6 | 0.0011 | 0.1200 | 0.6661         | 0.0001        |

### 7.2 Effect of Rho on Design Rankings (DIM)

#### MSE Dynamics for Design 1

Incidence: iid Uniform

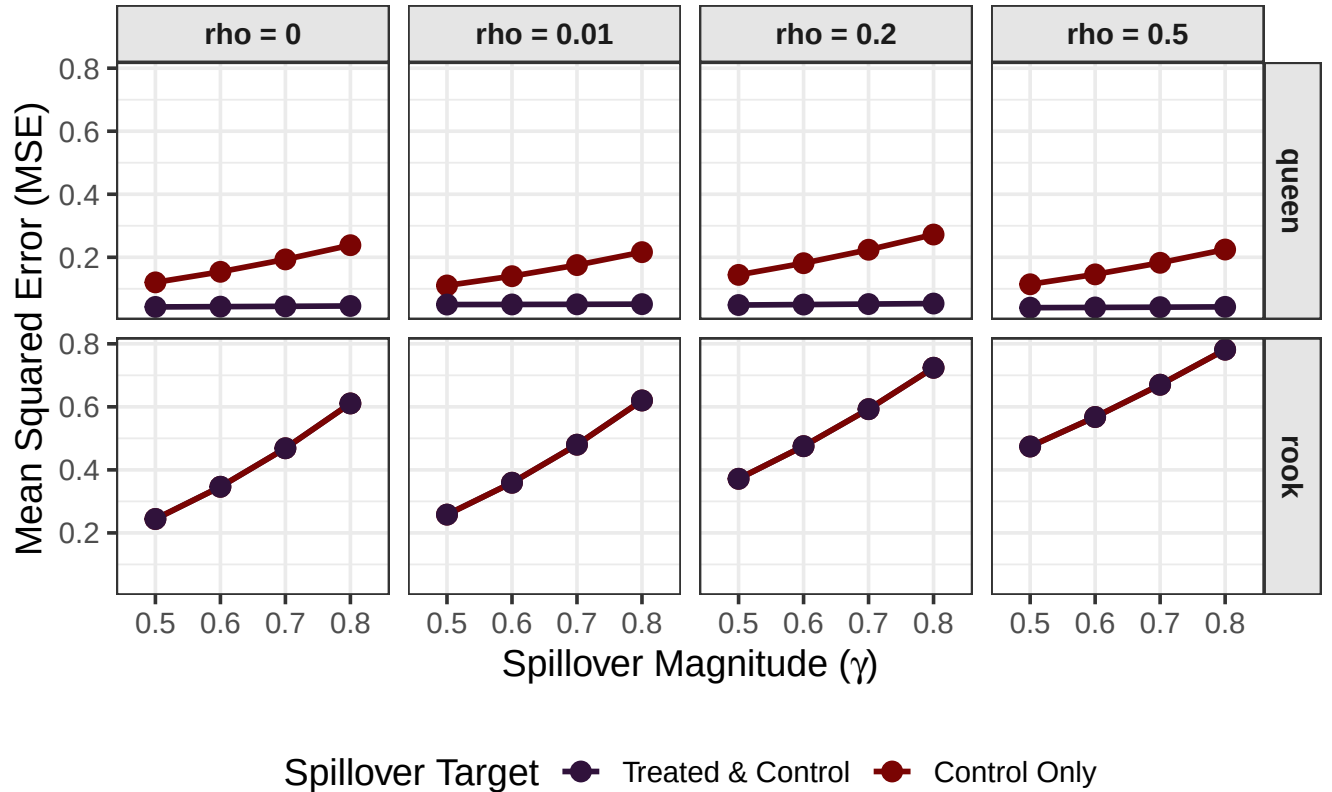


Figure 9: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

## MSE Dynamics for Design 2

Incidence: iid Uniform

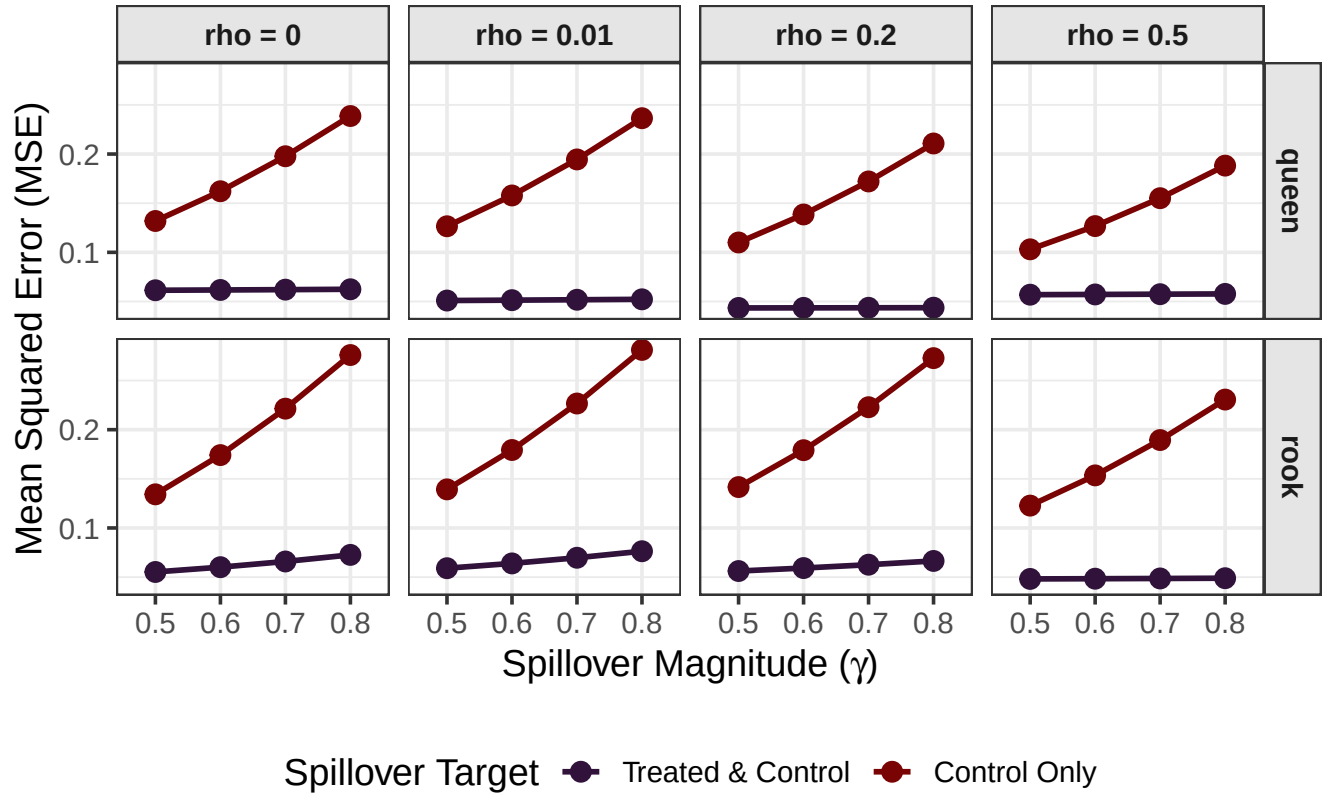


Figure 10: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

## MSE Dynamics for Design 3

Incidence: iid Uniform

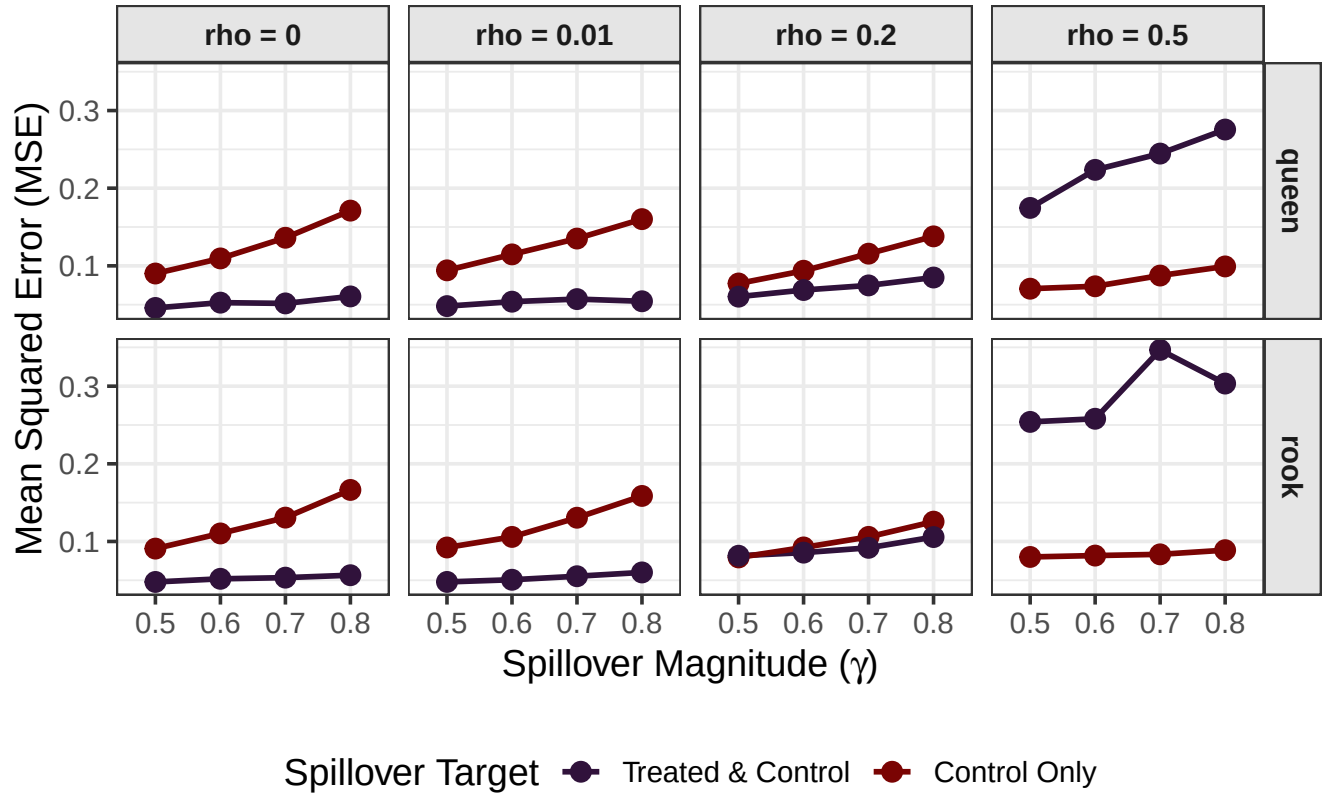


Figure 11: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

## MSE Dynamics for Design 4

Incidence: iid Uniform

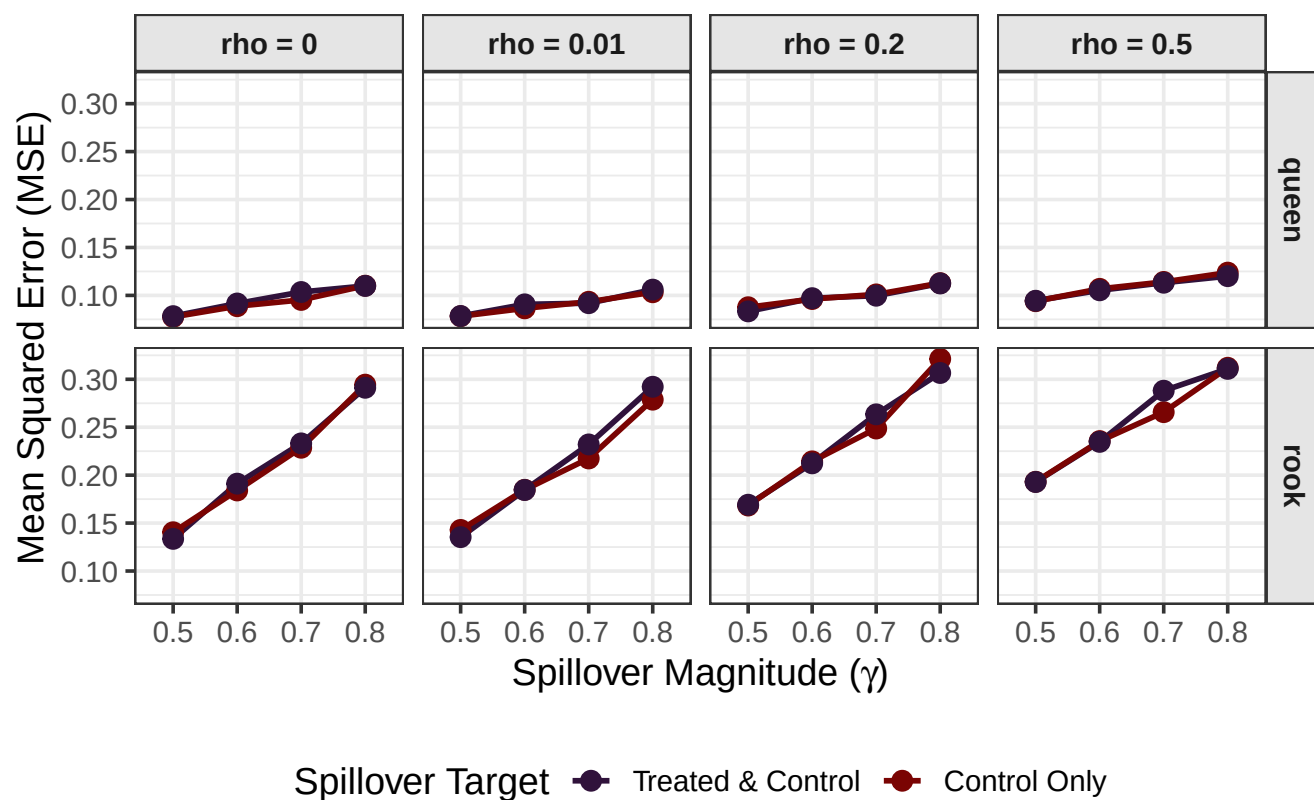


Figure 12: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

## MSE Dynamics for Design 5

Incidence: iid Uniform

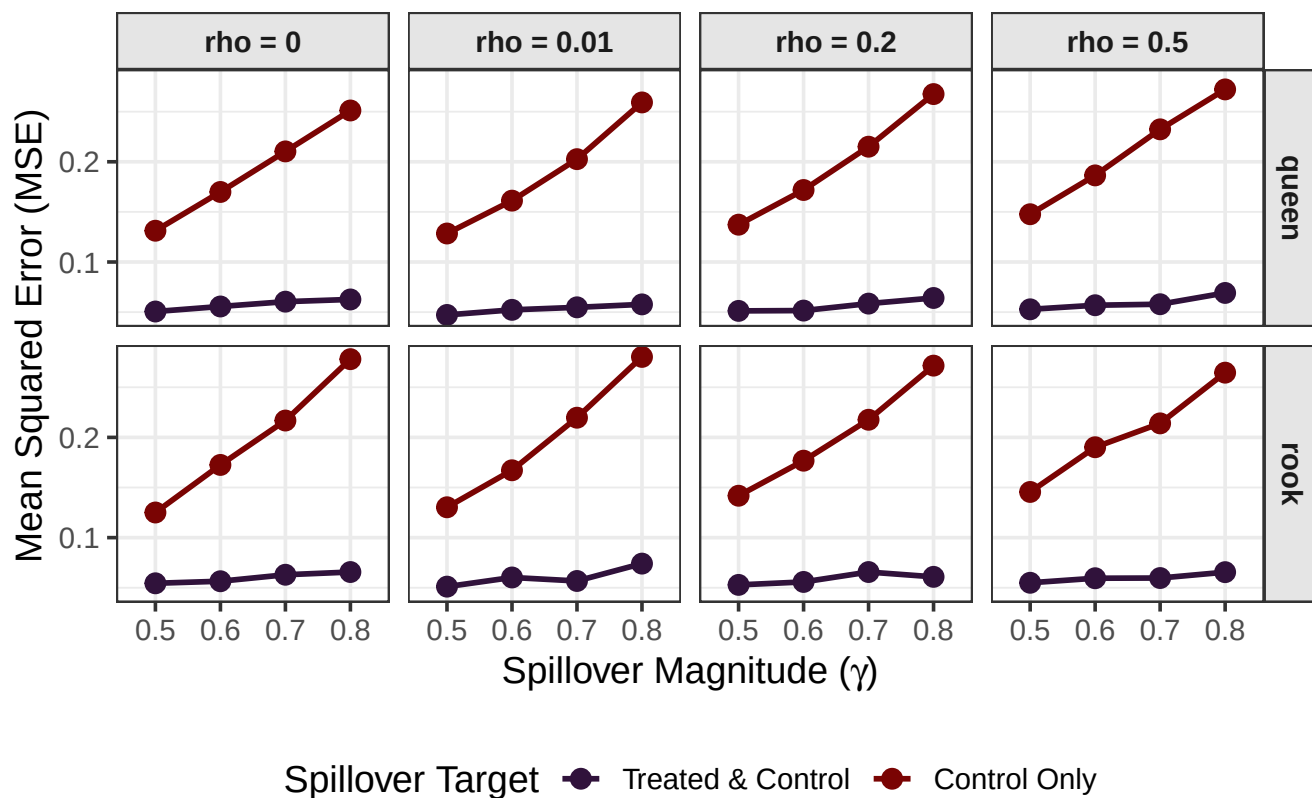


Figure 13: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

## MSE Dynamics for Design 6

Incidence: iid Uniform

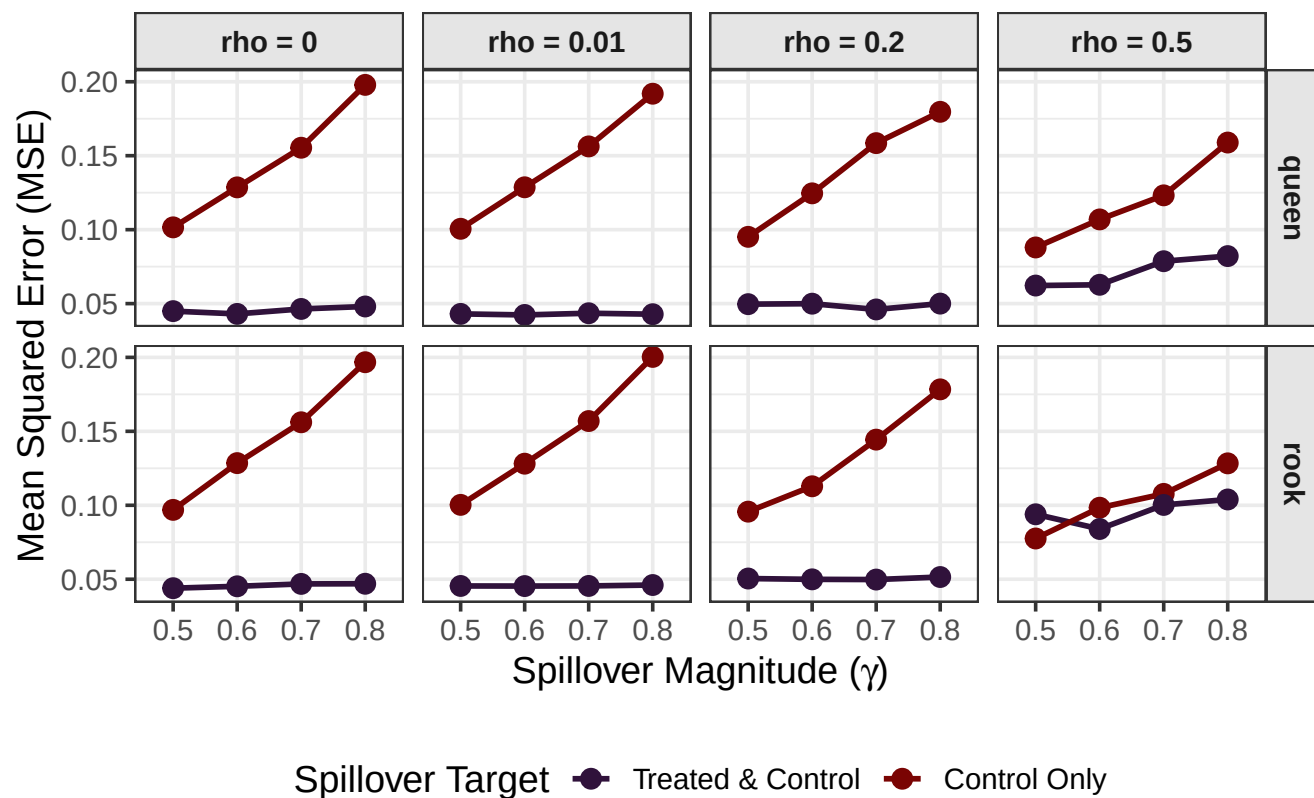


Figure 14: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

## \$`Design 1`

## MSE Dynamics for Design 1

Incidence: iid Uniform

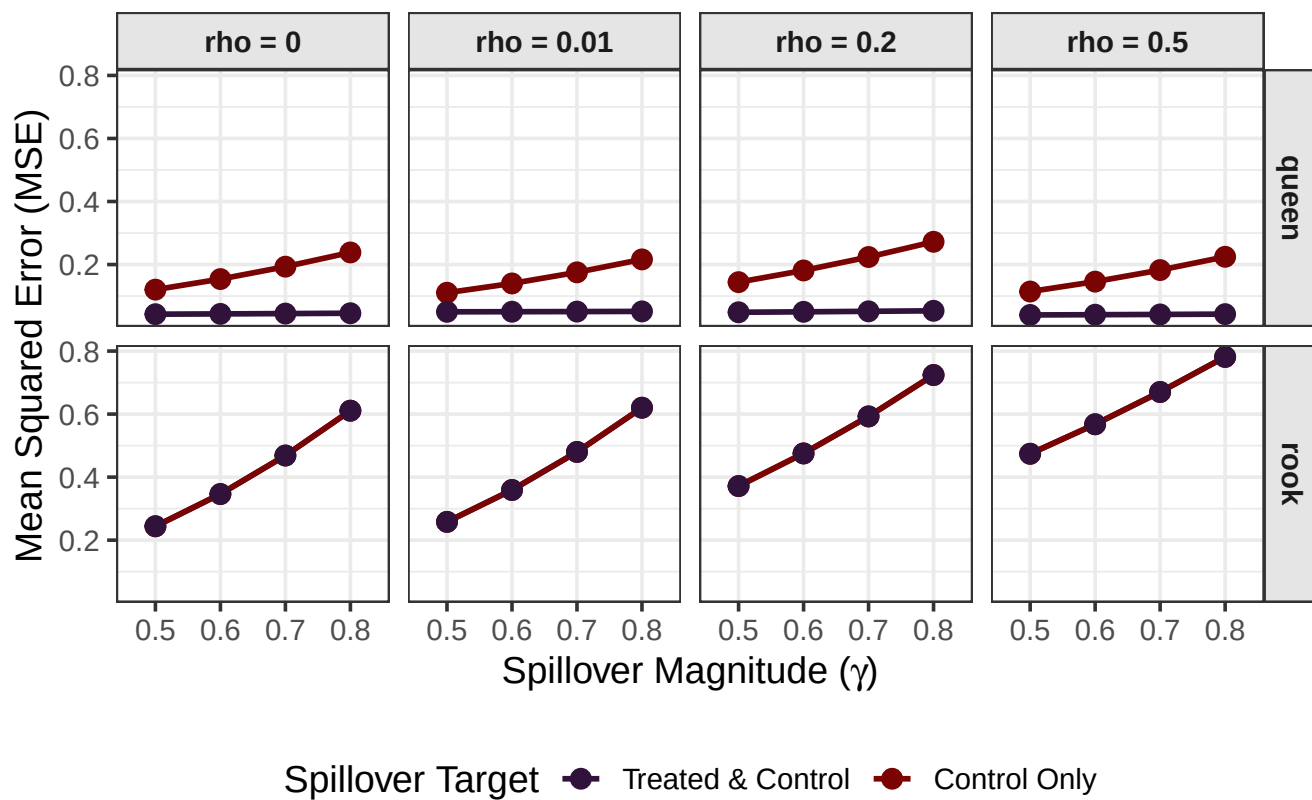


Figure 15: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 2`
```

## MSE Dynamics for Design 2

Incidence: iid Uniform

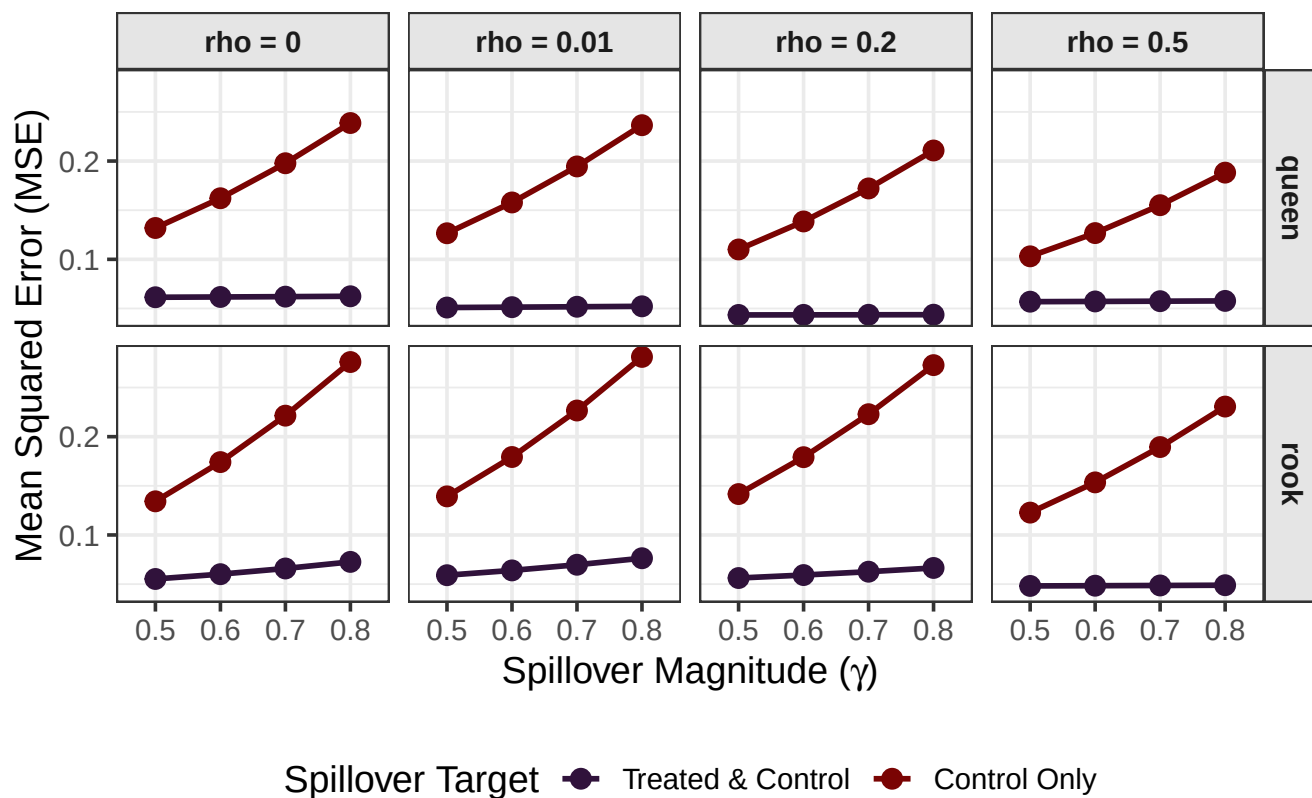


Figure 16: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 3`
```

## MSE Dynamics for Design 3

Incidence: iid Uniform

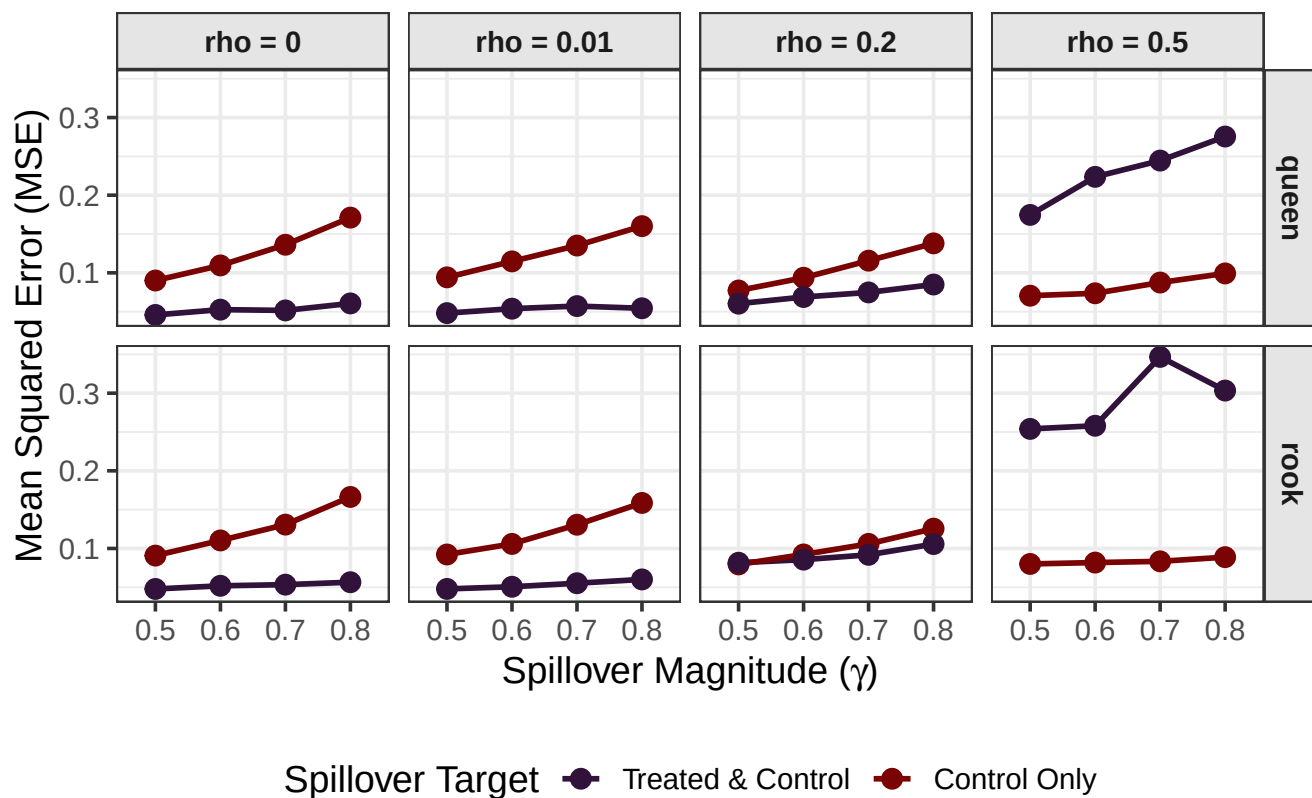


Figure 17: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 4`
```

## MSE Dynamics for Design 4

Incidence: iid Uniform

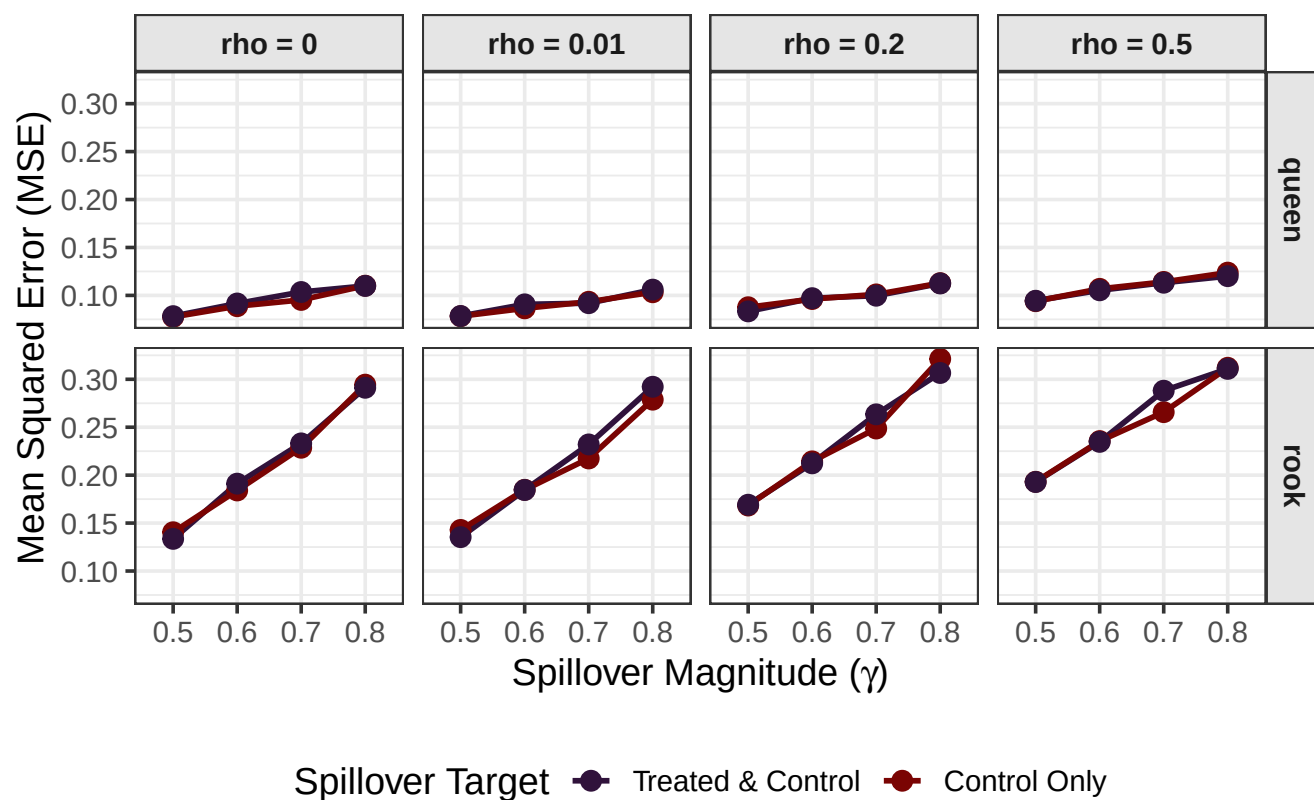


Figure 18: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 5`
```

## MSE Dynamics for Design 5

Incidence: iid Uniform

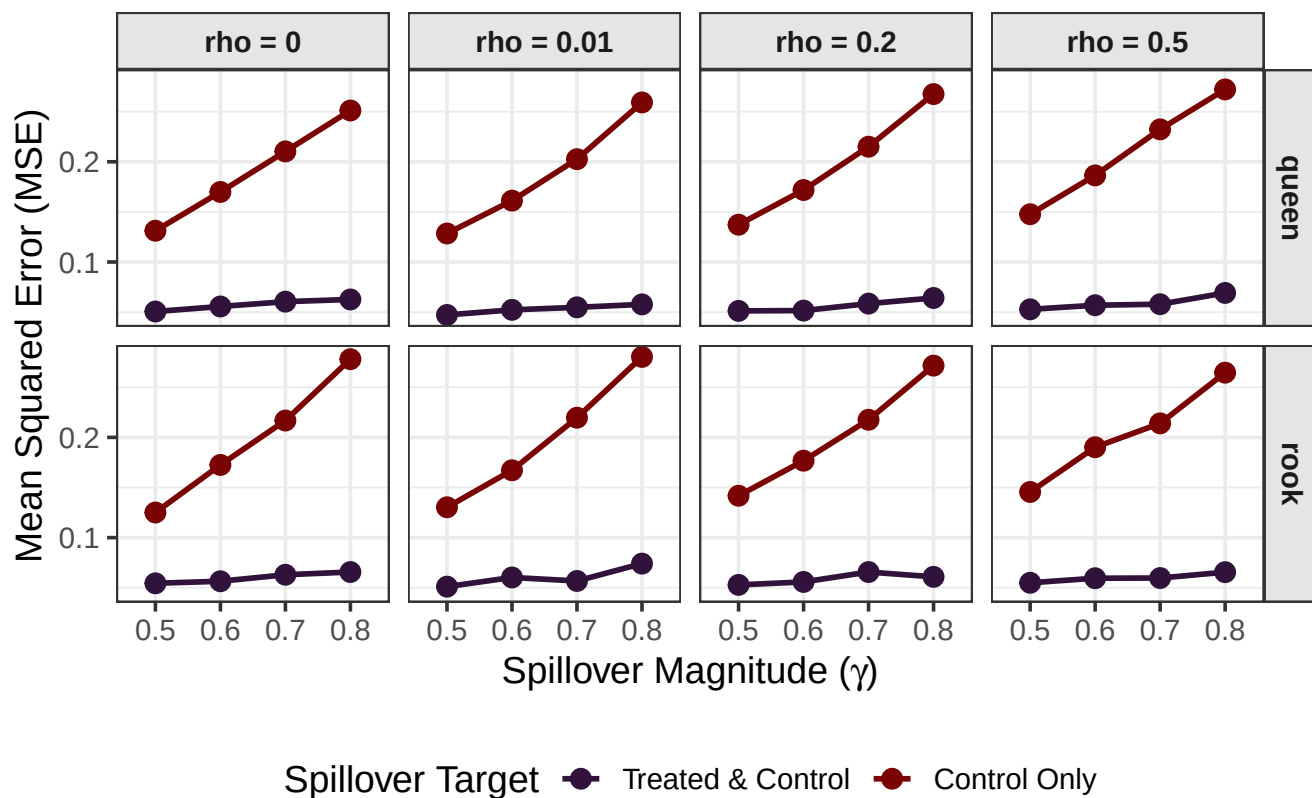


Figure 19: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 6`
```

## MSE Dynamics for Design 6

Incidence: iid Uniform

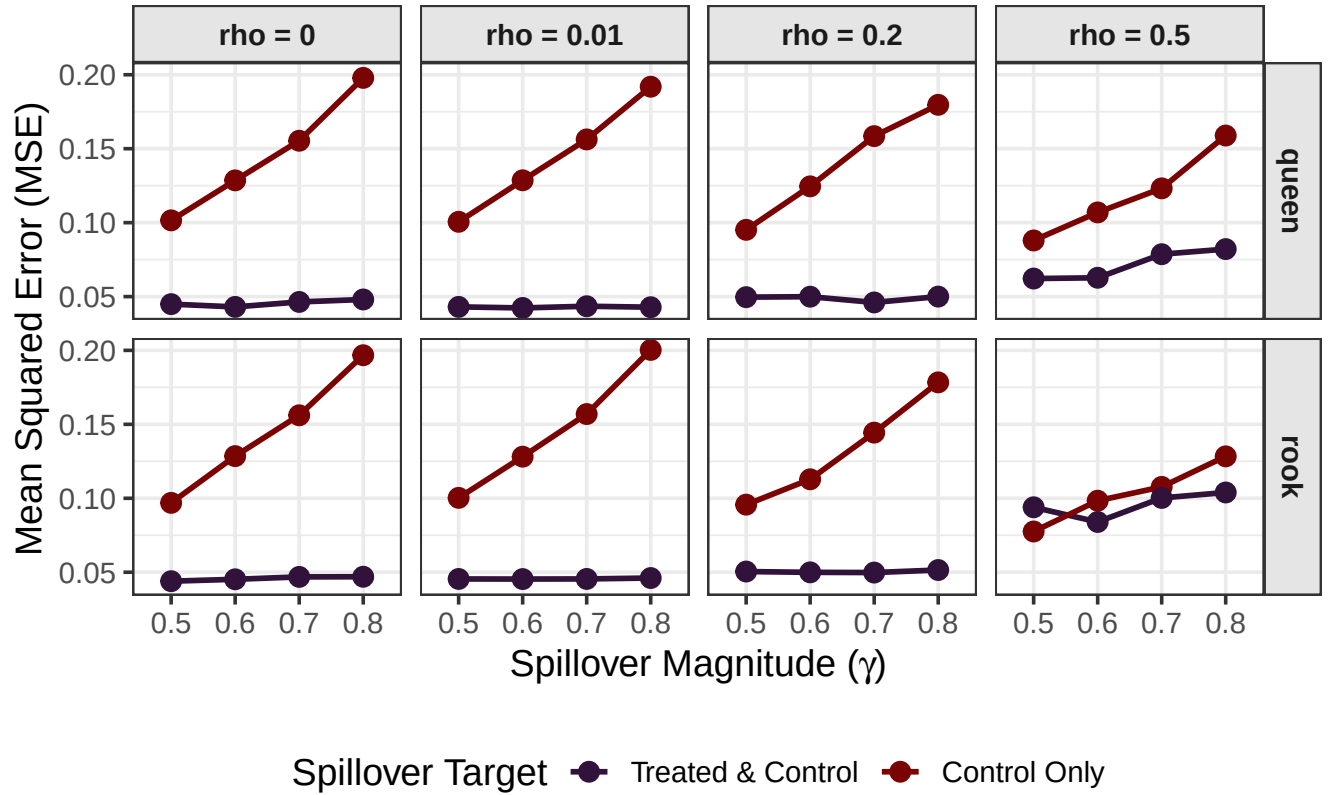


Figure 20: DIM (pre-revision): Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

### 7.3 Incidence Mode Comparison

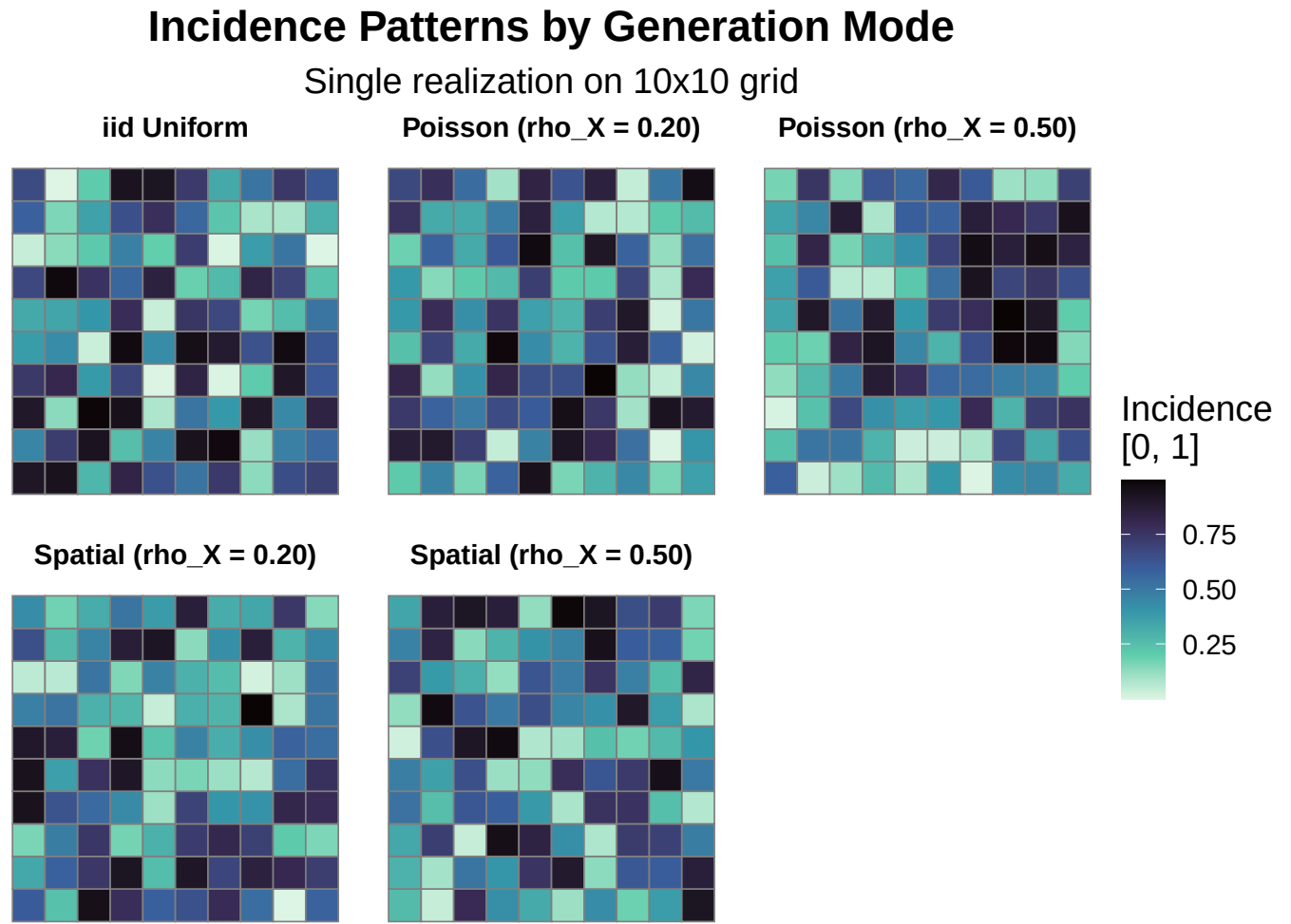


Figure 21: Incidence spatial patterns across the three generation modes for  $\rho_X = 0.20$  and  $0.50$ . Color represents incidence (darker = higher burden). Spatial and Poisson modes show visible spatial clustering absent from iid.

## Marginal Distribution of Incidence Values by Mode

Pooled across 50 resamples on 10x10 grid

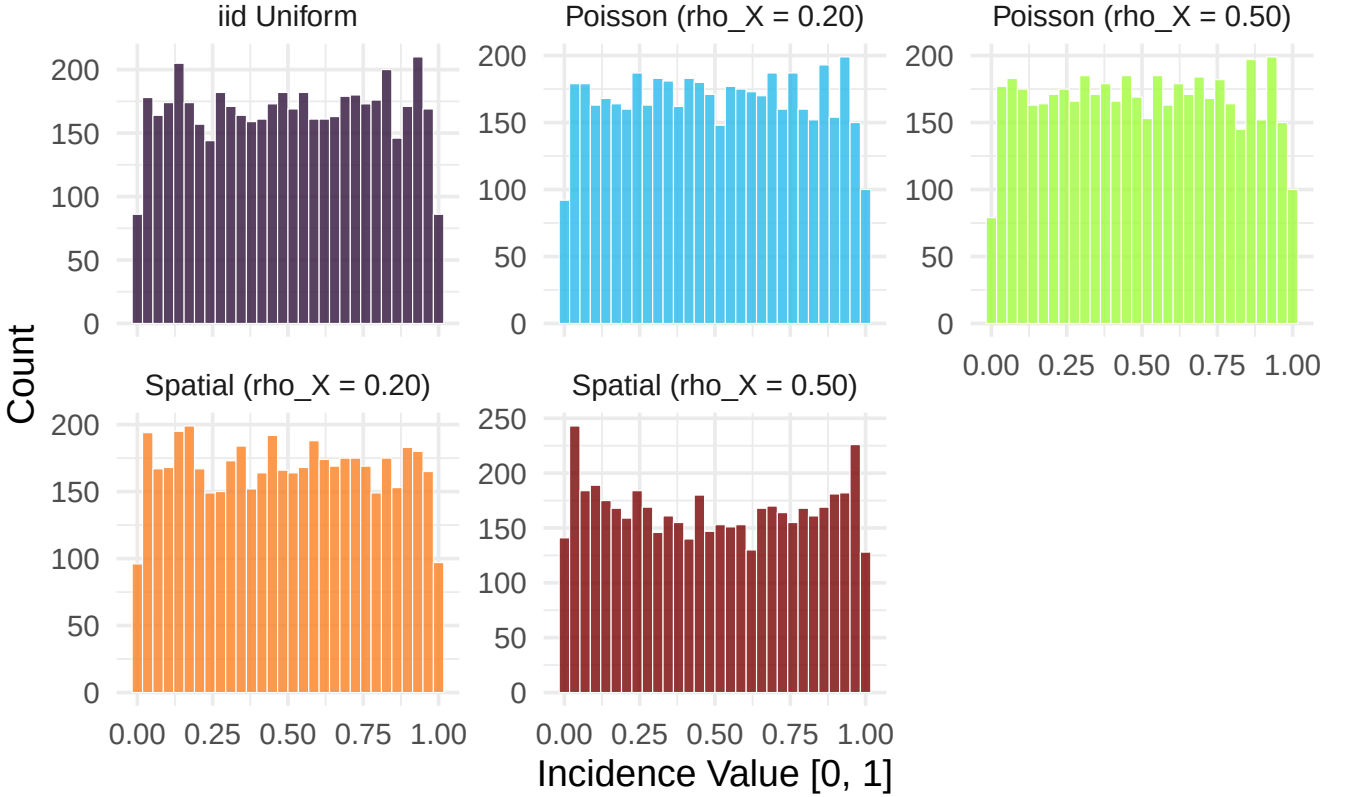


Figure 22: Marginal distribution of incidence values by mode, pooled across 50 resamples on the 10x10 grid. All modes are marginally Uniform on  $[0,1]$ ; spatial structure manifests in spatial covariance, not the marginal.

## 8 Subgroup Results by Incidence Mode

The following tables report DIM (pre-revision baseline) results separately for each incidence configuration, pooled over rook and queen. Results are **never aggregated across incidence modes**, as each represents a fundamentally different assumption about prior data.

### 8.1 DIM: iid Uniform

Table 12: DIM (pre-revision) robustness profile: iid Uniform

| Design   | Best_MSE | Q25_MSE | Median_MSE | Q75_MSE | Worst_MSE | IQR_MSE |
|----------|----------|---------|------------|---------|-----------|---------|
| Design 2 | 0.0434   | 0.0574  | 0.0897     | 0.1754  | 0.2812    | 0.1180  |
| Design 3 | 0.0458   | 0.0605  | 0.0904     | 0.1306  | 0.3467    | 0.0701  |
| Design 6 | 0.0423   | 0.0478  | 0.0945     | 0.1284  | 0.2003    | 0.0806  |
| Design 5 | 0.0472   | 0.0575  | 0.0996     | 0.1933  | 0.2803    | 0.1358  |

|          |        |        |        |        |        |        |
|----------|--------|--------|--------|--------|--------|--------|
| Design 4 | 0.0775 | 0.0969 | 0.1287 | 0.2295 | 0.3213 | 0.1326 |
| Design 1 | 0.0401 | 0.0962 | 0.2441 | 0.4764 | 0.7815 | 0.3801 |

## 8.2 DIM: Poisson ( $\rho_X = 0.20$ )

Table 13: DIM (pre-revision) robustness profile: Poisson ( $\rho_X = 0.20$ )

| Design   | Best_MSE | Q25_MSE | Median_MSE | Q75_MSE | Worst_MSE | IQR_MSE |
|----------|----------|---------|------------|---------|-----------|---------|
| Design 2 | 0.0447   | 0.0569  | 0.0706     | 0.0870  | 0.1132    | 0.0301  |
| Design 3 | 0.0467   | 0.0620  | 0.0868     | 0.1307  | 0.3198    | 0.0686  |
| Design 6 | 0.0432   | 0.0472  | 0.0905     | 0.1292  | 0.1990    | 0.0819  |
| Design 5 | 0.0487   | 0.0568  | 0.1002     | 0.1890  | 0.2745    | 0.1322  |
| Design 4 | 0.0792   | 0.0975  | 0.1343     | 0.2251  | 0.3123    | 0.1276  |
| Design 1 | 0.0434   | 0.0964  | 0.2541     | 0.4971  | 0.7967    | 0.4007  |

## 8.3 DIM: Poisson ( $\rho_X = 0.50$ )

Table 14: DIM (pre-revision) robustness profile: Poisson ( $\rho_X = 0.50$ )

| Design   | Best_MSE | Q25_MSE | Median_MSE | Q75_MSE | Worst_MSE | IQR_MSE |
|----------|----------|---------|------------|---------|-----------|---------|
| Design 2 | 0.0538   | 0.0716  | 0.0836     | 0.1076  | 0.3393    | 0.0360  |
| Design 3 | 0.0480   | 0.0638  | 0.0883     | 0.1318  | 0.3915    | 0.0680  |
| Design 6 | 0.0427   | 0.0466  | 0.0902     | 0.1319  | 0.2000    | 0.0854  |
| Design 5 | 0.0456   | 0.0571  | 0.0990     | 0.1910  | 0.2739    | 0.1339  |
| Design 4 | 0.0795   | 0.0993  | 0.1334     | 0.2189  | 0.3146    | 0.1196  |
| Design 1 | 0.0319   | 0.0804  | 0.2761     | 0.5354  | 0.7952    | 0.4550  |

## 8.4 DIM: Spatial ( $\rho_X = 0.20$ )

Table 15: DIM (pre-revision) robustness profile: Spatial ( $\rho_X = 0.20$ )

| Design   | Best_MSE | Q25_MSE | Median_MSE | Q75_MSE | Worst_MSE | IQR_MSE |
|----------|----------|---------|------------|---------|-----------|---------|
| Design 3 | 0.0472   | 0.0675  | 0.0891     | 0.1361  | 0.3716    | 0.0687  |
| Design 6 | 0.0436   | 0.0468  | 0.0906     | 0.1270  | 0.2023    | 0.0802  |
| Design 2 | 0.0415   | 0.0511  | 0.0941     | 0.1375  | 0.2165    | 0.0865  |
| Design 5 | 0.0481   | 0.0593  | 0.1019     | 0.1877  | 0.2834    | 0.1284  |
| Design 4 | 0.0776   | 0.0963  | 0.1293     | 0.2338  | 0.3097    | 0.1374  |
| Design 1 | 0.0324   | 0.0850  | 0.2532     | 0.5083  | 0.7986    | 0.4234  |

## 8.5 DIM: Spatial ( $\rho_X = 0.50$ )

Table 16: DIM (pre-revision) robustness profile: Spatial ( $\rho_X = 0.50$ )

| Design   | Best_MSE | Q25_MSE | Median_MSE | Q75_MSE | Worst_MSE | IQR_MSE |
|----------|----------|---------|------------|---------|-----------|---------|
| Design 2 | 0.0396   | 0.0667  | 0.0879     | 0.1258  | 0.3491    | 0.0591  |
| Design 6 | 0.0417   | 0.0472  | 0.0920     | 0.1280  | 0.2029    | 0.0808  |
| Design 3 | 0.0481   | 0.0688  | 0.0927     | 0.1299  | 0.3553    | 0.0610  |
| Design 5 | 0.0474   | 0.0561  | 0.1023     | 0.1911  | 0.2841    | 0.1350  |
| Design 4 | 0.0774   | 0.0982  | 0.1343     | 0.2289  | 0.3254    | 0.1308  |

## 9 Tau Sensitivity Analysis

The tau-sweep simulation (12,800 scenarios) swept the direct treatment effect  $\tau$  across  $\{0.8, 1.0, 1.5, 2.0, 3.0\}$ . The primary scenario is  $\tau = 1.0$ ; all standard results above use this slice. This section examines whether design rankings and statistical power change across effect sizes. Plots show iid Uniform incidence under queen contiguity (primary). All  $\tau$  levels reuse the same assignments and noise (common random numbers), so agreement across  $\tau$  is not independent confirmation.

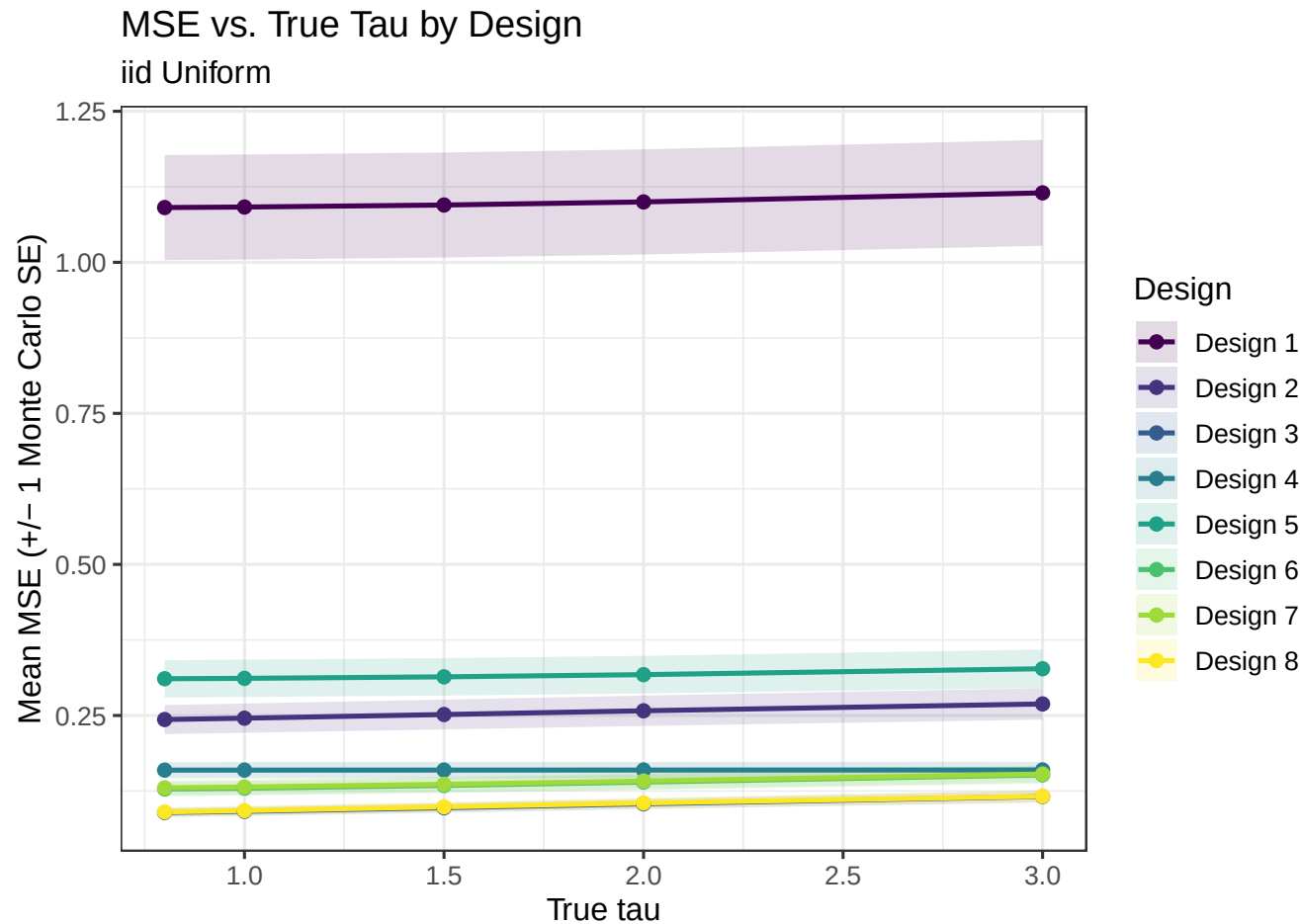


Figure 23: MLE MSE by design across tau values (iid Uniform, queen). Checkerboard has the highest mean MSE at every tau.

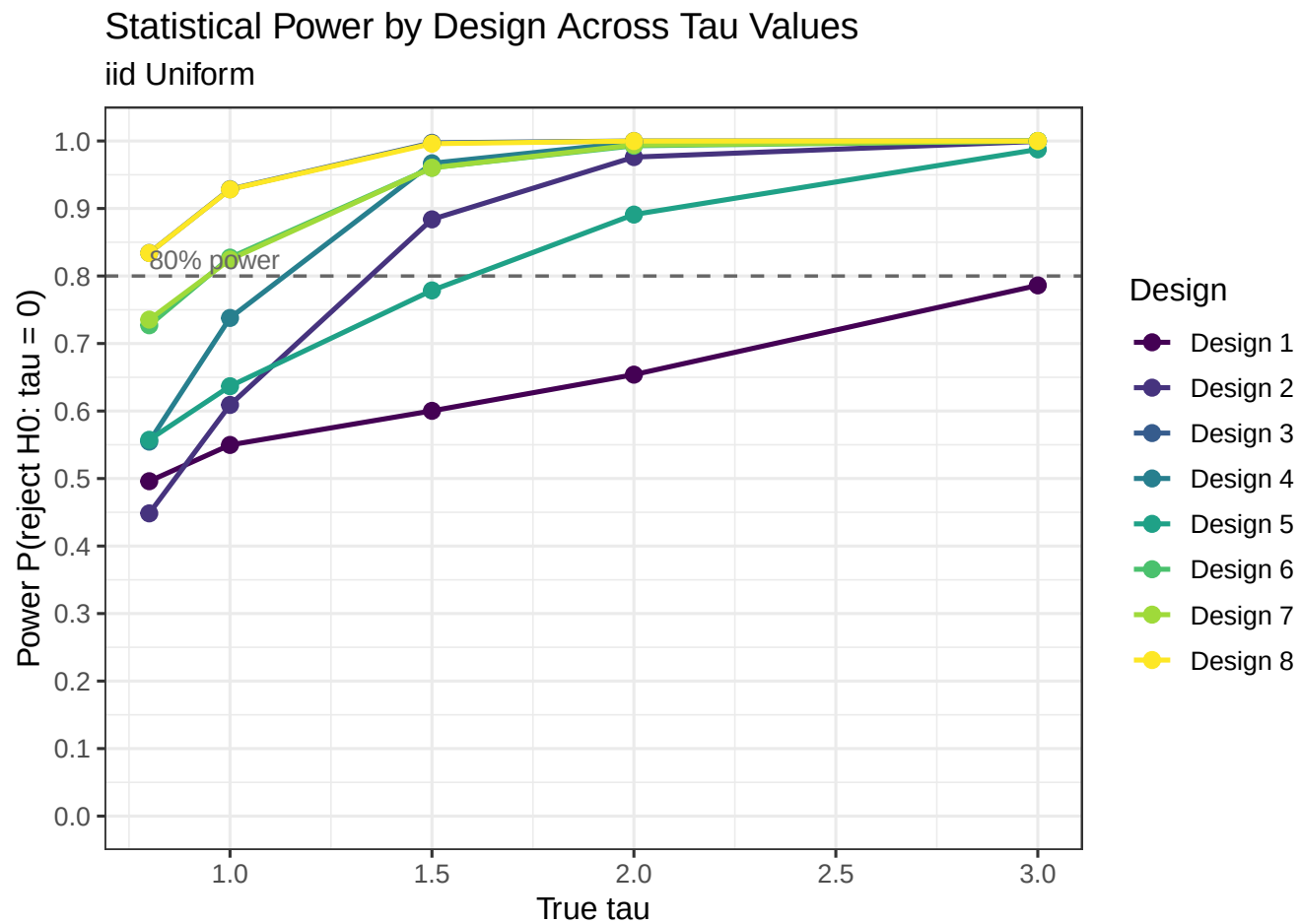


Figure 24: Statistical power  $P(\text{reject } H_0: \tau=0)$  by design across  $\tau$  values (iid Uniform, queen). 80% power reference line shown. Designs with higher MSE require larger  $\tau$  to achieve adequate power.

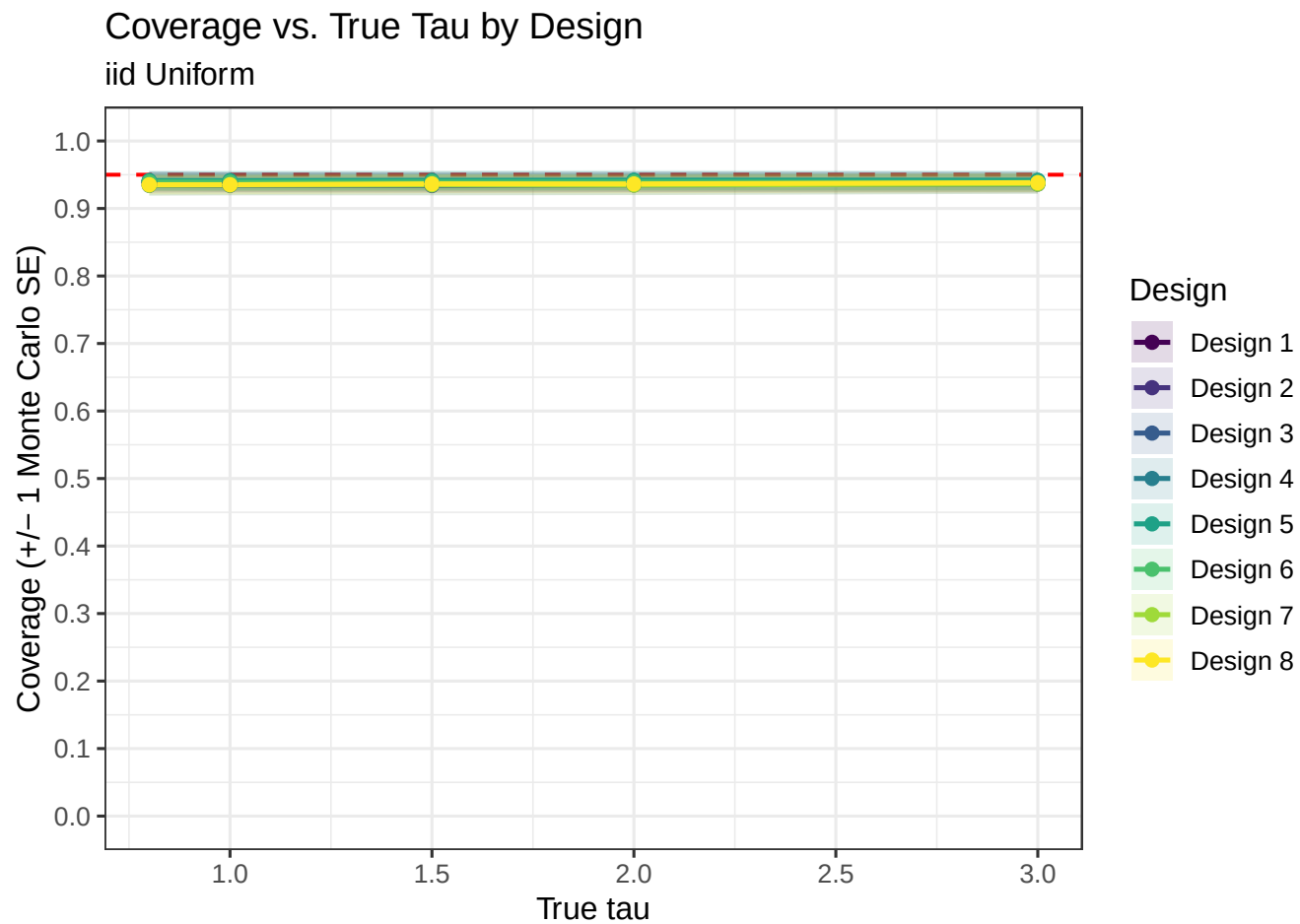


Figure 25: MLE coverage by design across tau values (iid Uniform, queen). Nominal 95% reference line shown. Coverage is near-nominal for every design, including Checkerboard; under rook (not shown) Checkerboard is not identified.



Figure 26: Design rank trajectories across tau values (iid Uniform, queen). Rank 1 = best MSE. Stable trajectories indicate robust design recommendations across effect sizes.

## 10 Conclusions

### 10.1 Summary of Findings

The simulation study evaluated 8 treatment assignment designs across 2,560 MLE parameter scenarios (primary  $\tau = 1.0$  slice), spanning three incidence generation modes, two contiguity structures, four spatial autocorrelation levels, four spillover magnitudes, and two spillover types. The full tau-sweep covers 12,800 scenarios across  $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ .

**Key findings** (MLE,  $\tau = 1.0$ ; queen primary, rook sensitivity):

1. **Coverage is near-nominal and does not separate the designs.** Mean coverage is 0.936–0.942 for every design under both neighbor types, except Checkerboard under rook. The design differences show up in MSE. The pre-revision DIM baseline under-covers (mean 0.72), but it was run under the old data-generating process and is not a like-for-like comparison.
2. **Saturation Quadrants and Incidence-Guided Saturation Quadrants have the lowest MSE** (queen: 0.090 and 0.091; rook: 0.071 and 0.072). The two are statistically indistinguishable (queen Nemenyi  $p = 0.985$ , Holm-adjusted Wilcoxon  $p = 0.413$ ; eight-design comparisons in

results/eight\_design\_supplementary/).

3. **Balanced Halves and Balanced Quartiles form the next tier** (queen: 0.130 and 0.131), and are also statistically indistinguishable (queen Nemenyi  $p = 1.000$ , Wilcoxon  $p = 0.884$ ). Isolation Buffer (0.160), High Incidence Focus (0.245) and 2x2 Blocking (0.319) follow under queen; under rook, 2x2 Blocking (0.128) and Isolation Buffer (0.133) are close and High Incidence Focus (0.206) is higher.
4. **Checkerboard has the highest MSE.** Under queen its MSE is 1.087, with mean bias 0.001, so almost all of it is variance. Under rook,  $WZ = \mathbf{1} - Z$  and  $\tau$  is not identified (bias -0.652, approximately  $-\gamma$ ; coverage 0.150); these estimates are kept but flagged.
5. **The MSE ordering is the same at every  $\tau$  under queen** ( $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ ), with the caveat that all  $\tau$  levels share the same assignments and noise.

## 10.2 Limitations and Planned Extensions

- **Oracle MLE:** The primary MLE model uses the true spillover covariate. A non-oracle fit ( $Y \sim Z + X$ ) was run as a sensitivity analysis (`sim_results_MLEnonoracle_tau_sweep_combined_20260924_025509.rds`) but is not shown here.
- **DIM baseline is pre-revision:** DIM (2,500 fits per scenario) was run under the old data-generating process with 6 designs and was not re-run for the 2026-09 revision.
- **Equal populations:** Poisson mode uses  $P_i = 100,000$  for all clusters. Heterogeneous population sizes (log-normal) would more closely match NC county data.
- **Grid sensitivity:** All results use  $G = 10$  ( $N = 100$ ). Sensitivity to  $G = 8$  or  $G = 15$  has not been evaluated.
- **Tau sensitivity:**  $\tau$  was swept across  $\{0.8, 1.0, 1.5, 2.0, 3.0\}$  in the completed tau-sweep run (12,800 scenarios). See Section 9 for power and rank-stability results across effect sizes.

## 11 References

- Mirzaei, A. et al. (2019). Sudden unexpected death rates. (*SUD base rate 35/100,000*)
- Gan, W. et al. (2019). County-level Poisson regression for SUD mortality.
- Watson, K. et al. (AHA). Census tract-level spatial analysis of SUD.
- LeSage, J. & Pace, R.K. (2009). *Introduction to Spatial Econometrics*. CRC Press.
- Bivand, R., et al. `spdep` R package.
- Bivand, R. & Piras, G. `spatialreg` R package.
- R packages used: `dplyr`, `tidyr`, `ggplot2`, `viridis`, `knitr`, `kableExtra`, `digest`

## 12 Appendix: Result File Locations

Table 17: Key output files in `results/sim_data/` and `results/` directories.

| File                                                                             | Contents                                                                 |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <code>sim_results_MLE_tau_sweep_combined_20260924_025509.rds</code>              | PRIMARY: 12,800 oracle MLE tau-sweep scenarios (2026-09 revision)        |
| <code>sim_results_MLE_tau_sweep_{iid,spatial,poisson}_20260924_025509.rds</code> | Per-incidence-mode splits of the oracle results                          |
| <code>sim_results_MLEnonoracle_tau_sweep_combined_20260924_025509.rds</code>     | Non-oracle MLE ( $Y \sim Z + X$ ) sensitivity analysis, 12,800 scenarios |
| <code>surface_results_MLE_tau_sweep_20260924_025509.rds</code>                   | One row per (scenario, surface); basis of the Monte Carlo SEs            |
| <code>sim_results_DIM_combined_20260304_195321.rds</code>                        | DIM baseline: 1,920 scenarios (pre-revision DGP, 6-design sweep)         |
| <code>MLE_tau_sweep_design_recommendations.pdf</code>                            | Design recommendation figures + tables (primary tau=1.0)                 |
| <code>MLE_tau_sweep_incidence_overview.pdf</code>                                | Incidence heatmaps and distributions                                     |
| <code>mle_per_config/MLE_tau_sweep_*.pdf</code> (5 files, per incidence config)  | Per-config plot PDFs (primary tau=1.0 slice)                             |
| <code>mle_per_config/MLE_tau_sweep_*_tau_sensitivity.pdf</code> (5 files)        | Per-config tau sensitivity PDFs (MSE/power/coverage vs tau)              |

|                                              |                                                                |
|----------------------------------------------|----------------------------------------------------------------|
| 11_statistical_comparisons_report.{html,pdf} | Formal statistical comparisons (Friedman, Nemenyi, Wilcoxon)   |
| 00_mathematical_specification.pdf            | Mathematical theory document                                   |
| 07_results_summary.pdf                       | This report                                                    |
| archive/pre_revision_20260924/               | Superseded pre-revision (March/April 2026) MLE results and rep |

All simulation data files are located in **results/sim\_data/**. All rendered reports and PDFs are in **results/** and **paper/report/**.